

Cholesterol taste avoidance in *Drosophila melanogaster*

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This **useful** study provides **convincing** evidence that *Drosophila* can taste cholesterol through a subset of bitter-sensing gustatory receptor neurons, and that flies avoid high-cholesterol food. However, the same receptors have been previously found to be involved in the detection of multiple seemingly unrelated chemicals, and the reported expression patterns of these receptors contradict past reports. These caveats are not mentioned in the paper, raising critical concerns about the study's conclusions.

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Abstract

The question as to whether animals taste cholesterol taste is not resolved. This study investigates whether the fruit fly, *Drosophila melanogaster*, is capable of detecting cholesterol through their gustatory system. We found that flies are indifferent to low levels of cholesterol and avoid higher levels. The avoidance is mediated by gustatory receptor neurons (GRNs), demonstrating that flies can taste cholesterol. The cholesterol responsive GRNs comprise a subset that also respond to bitter substances. Cholesterol detection depends on five ionotropic receptor (IR) family members, and disrupting any of these genes impairs the flies' ability to avoid cholesterol. Ectopic expressions of these IRs in GRNs reveals two classes of cholesterol receptors, each with three shared IRs and one unique subunit. Additionally, expressing cholesterol receptors in sugar-responsive GRNs confers attraction to cholesterol. This study reveals that flies can taste cholesterol, and that the detection depends on IRs in GRNs.

Introduction

Many types of tastants are beneficial at low concentrations, and harmful at high levels. Examples include minerals such as Na^+ and Ca^{2+} (Zhang, Ni et al. 2013 , Lee, Poudel et al. 2018 ) and fatty acid such as hexanoic acid (Ahn, Chen et al. 2017 , Pradhan, Shrestha et al. 2023 ). Organic molecules such as cholesterol are a crucial sterol that plays essential roles in cellular membrane integrity, signaling functions, and steroid hormone synthesis. This vital molecule supports

numerous biological processes in animals, including reproduction, nutrient transport, and cellular activation (Igarashi, Ogiwara et al. 2018). However, excessive cholesterol consumption can lead to a host of poor health consequences, including cardiovascular disease, and type 2 diabetes (Soliman 2018). Due to the bivalent impact of cholesterol on human health, it stands to reason that there may be mechanisms that exist to promote or repress the taste of cholesterol. However, it is not clear whether cholesterol is sensed by the mammalian taste system. Mice and humans express several dozen taste receptors, most of which function in bitter taste (referred to as either T2Rs or TAS2Rs). The activities of two of these bitter receptors, T2R4 and T2R14 have been shown to be modulated by cholesterol. However, it is unclear if they contribute to the taste of cholesterol (Pydi, Jafurulla et al. 2016, Shaik, Jagupilli et al. 2019, Kim, Gumpfer et al. 2024).

It is plausible that insects such as *Drosophila melanogaster* might display a gustatory attraction to low levels of cholesterol since unlike vertebrates, which can synthesize cholesterol internally, fruit flies must obtain sterols through their diet (Clark and Bloch 1959, Niwa and Niwa 2011, Shaheen 2020). Insects acquire cholesterol primarily from plant-derived phytosterols or pre-existing cholesterol (Jing and Behmer 2020). For instance, *Manduca sexta* and *Bombyx mori* convert plant sterols to cholesterol through dealkylation in their gut, which is essential for producing hormones such as ecdysone (Igarashi, Ogiwara et al. 2018). *Drosophila* acquires cholesterol directly from dietary sources such as phytosterols (e.g., sitosterol, stigmasterol) and fungal sterols (e.g., ergosterol) found in yeast (Niwa and Niwa 2011). Given that consumption of high cholesterol is harmful (Soliman 2018, Schade, Shey et al. 2020), fruit flies might display a gustatory aversion to high levels, while finding low levels attractive. Such a bivalent response would be similar to the flies' taste attraction to low concentrations of Na^+ and their repulsion to high Na^+ (Zhang, Ni et al. 2013, Jaeger, Stanley et al. 2018, Taruno and Gordon 2023, Sang, Dhakal et al. 2024). Ca^{2+} is also required at low levels and deleterious at high concentrations. However, we have previously shown that fruit flies are indifferent to modest levels of Ca^{2+} and avoid high Ca^{2+} (Lee, Poudel et al. 2018). Thus, if flies are endowed with the capacity to taste cholesterol, it is open question as to whether they would have a bivalent gustatory response depending on concentration, similar to Na^+ (Zhang, Ni et al. 2013, Jaeger, Stanley et al. 2018, Taruno and Gordon 2023, Sang, Dhakal et al. 2024), or be indifferent to low cholesterol and reject high cholesterol, similar to the flies' differential reaction to Ca^{2+} depending on concentration (Lee, Poudel et al. 2018).

In *Drosophila*, gustatory organs are distributed on multiple body parts, including the labellum at the end of the proboscis, which represents the largest taste organ. The labellum consists of two halves, each of which is decorated with 31 external bristles. These sensilla house either two or four gustatory receptor neurons (GRNs), which allow them to respond to external chemical stimulation and modulate behavioral responses (Dahanukar, Foster et al. 2001, Larsson, Domingos et al. 2004, Suh, Wong et al. 2004, Benton, Vannice et al. 2009, Cameron, Hiroi et al. 2010, Chen, Wang et al. 2010, Kim, Lee et al. 2010, Kwon, Kim et al. 2010, Rimal and Lee 2018). This is accomplished through expression of a diverse repertoire of receptor classes, including gustatory receptors (GRs), ionotropic receptors (IRs), pickpocket (PPK) ion channels, and transient receptor potential (TRP) channels.

In this work, we reveal that flies taste cholesterol. Reminiscent of their reaction to Ca^{2+} (Lee, Poudel et al. 2018), they are indifferent to low cholesterol and reject high cholesterol. Using a combination of behavioral and electrophysiological assays, we demonstrate that a subset of the same class of GRNs that respond to bitter chemicals is also required in adults for avoiding the taste of higher cholesterol levels. In addition, we found that multiple members of the IR family are involved in cholesterol taste perception, and that there are two overlapping sets of IRs that are sufficient to confer cholesterol sensitivity to GRNs that normally do not respond to cholesterol. This work establishes that flies can taste cholesterol, and defines the underlying cellular and molecular mechanisms involved in rejection of high cholesterol.

Results

Flies taste cholesterol through a subset of bitter GRNs

To address whether fruit flies can taste cholesterol, we investigated whether cholesterol triggers action potentials in GRNs associated with taste bristles in the labella. The 31 sensilla present in each hemisphere of the labellum are categorized into long (L), intermediate (I), and short (S) subtypes (**Figure 1A**) (Hiroi, Marion-Poll et al. 2002). To examine cholesterol-induced action potentials in response to a range of cholesterol concentrations, we focused on S7, I8, and L6 sensilla and performed tip recordings. We detected action potentials in S7 once the cholesterol concentration reached 10^{-3} %, whereas the I8 and L6 were nearly unresponsive even at 0.1% (**Figures 1B and C**). The methyl- β -cyclodextrin (M β CD) used to dissolve cholesterol did not evoke spikes (**Figure 1—figure supplement 1A and B**). We then analyzed all 31 sensilla using 0.1% cholesterol. We found that the S-type sensilla, especially the S6 and S7 sensilla, were most responsive, while very few spikes were induced from the I-type or L-type sensilla (**Figure 1D**).

To determine which GRN type contributes to cholesterol-induced action potentials, we selectively inactivated different classes of GRNs by expressing a transgene encoding the inwardly rectifying potassium channel, Kir2.1 (Baines, Uhler et al. 2001). The bristles on the labellum harbor GRNs that fall into four main classes, each of which expresses a gene driver unique to that class (Montell 2021). These include A GRNs, which respond to sugars, low Na^+ and other attractive compounds (*Gr64f-GAL4*), B GRNs, which are stimulated by bitter compounds, high Na^+ and other aversive chemicals (*Gr33a-GAL4*), C GRNs, which are activated by H_2O and hypo-osmolarity (*ppk28-GAL4*), and D GRNs, which respond to Ca^{2+} and high concentrations of other cations (*ppk23-GAL4*) (Thorne, Chromey et al. 2004, Dahanukar, Lei et al. 2007, Moon, Lee et al. 2009, Cameron, Hiroi et al. 2010, Lee, Poudel et al. 2018). We found that silencing B GRNs reduced neuronal responses to cholesterol, whereas inhibition of other GRN types exhibited normal neuronal firing (**Figures 1E and F**). These data demonstrate that cholesterol is sensed by B GRNs.

A cluster of IRs is required to sense cholesterol in adult *Drosophila*

To pinpoint the molecular sensors for detecting cholesterol, we first investigated requirements for the largest family of taste receptors—the GRs. Six GRs are broadly expressed in GRNs and three of them serve as co-receptors (Montell 2021, Shrestha and Lee 2023), including GR32a (Miyamoto and Amrein 2008), GR33a (Lee, Moon et al. 2009, Moon, Lee et al. 2009), GR39a.a (Dweck and Carlson 2020), GR66a (Moon, Kötting et al. 2006), GR89a (Shrestha and Lee 2021), and GR93a (Lee, Moon et al. 2009). We performed tip recordings, demonstrating that mutations disrupting any of these co-receptors had no impact on cholesterol-induced action potentials (**Figure 2—figure supplement 1A**). *Drosophila* encodes 13 TRP channels, several of which function in taste (Al-Anzi, Tracey et al. 2006, Kang, Pulver et al. 2010, Kim, Lee et al. 2010, Zhang, Raghuwanshi et al. 2013, Zhang, Raghuwanshi et al. 2013, Mandel, Shoaf et al. 2018, Leung, Thakur et al. 2020, Montell 2021). We analyzed mutant lines disrupting most of these channels and found that the neuronal responses were normal (**Figure 2—figure supplement 1B**).

IRs comprise another large family of receptors that function in taste, as well as in other sensory processes (Rytz, Croset et al. 2013, Rimal and Lee 2018). To address whether any IR is required for cholesterol taste, we screened the 32 available *Ir* mutants by performing tip recording on S7 sensilla using 0.1% cholesterol. Most mutants displayed normal responses (**Figure 2A**), including those with previously identified gustatory functions such as *Ir7a*¹ (acetic acid sensor) (Rimal, Sang et al. 2019), *Ir7c*^{GAL4} and *Ir60b*³ (high Na^+) (McDowell, Stanley et al. 2022, Sang, Dhakal et al. 2024), *Ir56b*¹ (low Na^+) (Dweck, Talross et al. 2022), *Ir62a*¹ (Ca^{2+}) (Lee, Poudel et al. 2018), *Ir94*^f (cantharidin) (Pradhan, Shrestha et al. 2024) as well as *Ir20a*¹, *Ir47a*¹, *Ir52a*¹, and *Ir92a*¹.

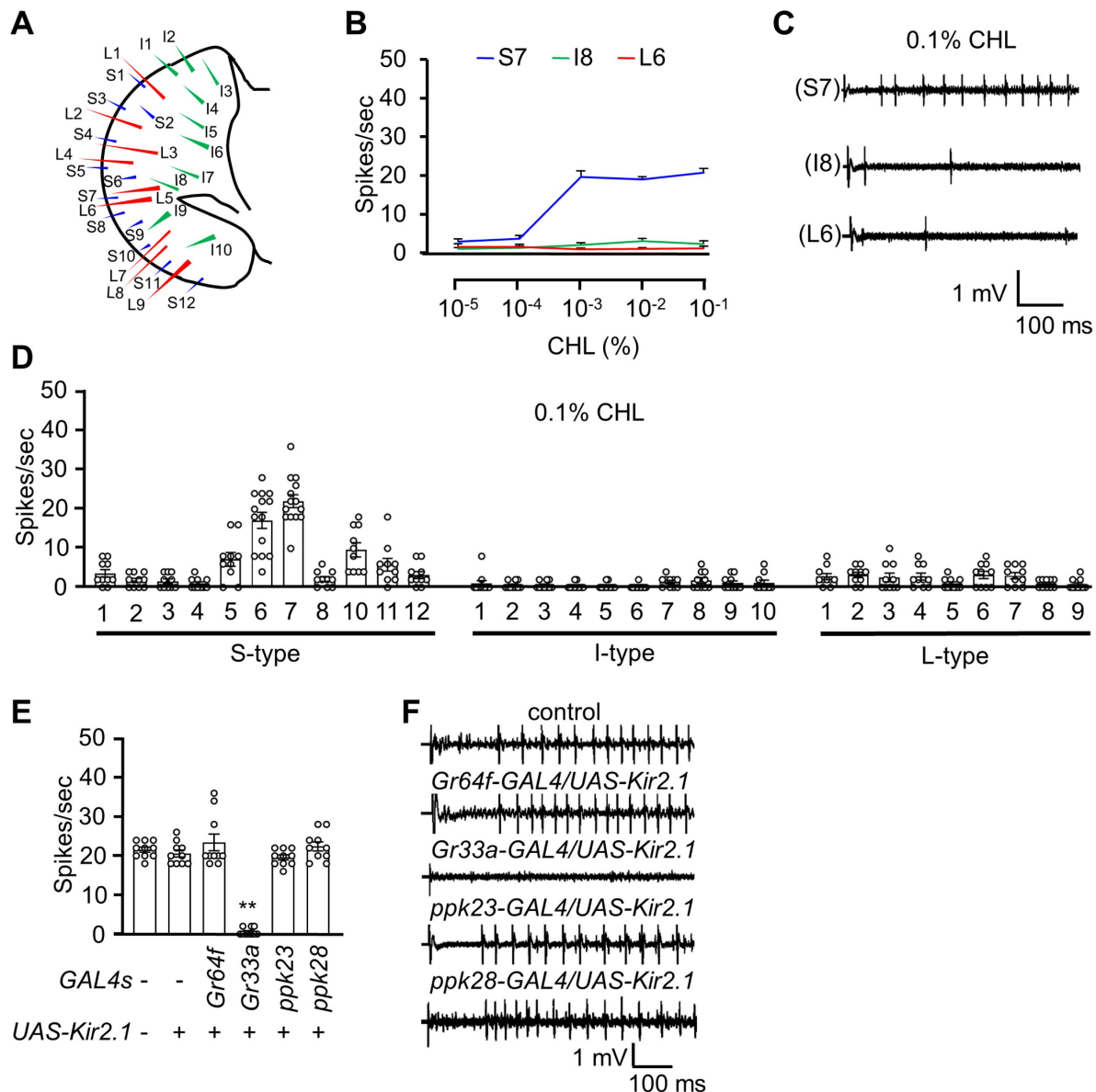


Figure 1.

The neuronal response of the adult flies to cholesterol.

(A) Schematic diagram of the fly labellum. (B) Average frequencies of action potential generated from S7, I8, and L6 sensilla upon the application of different concentrations of cholesterol ($n = 10-12$). (C) Representative sample traces of S7, I8, and L6 from (B). (D) Electrophysiological responses of control flies produced from all the labellum sensilla in response to 0.1% cholesterol ($n = 10-12$). (E) Electrophysiological analysis of the neuron-ablated flies by the inwardly rectifying potassium channel *Kir2.1* in presence of 0.1% cholesterol ($n = 10-12$). (F) Representative sample traces of the S7 sensilla from (E). All error bars represent SEM. Single-factor ANOVA was combined with Scheffe's post hoc analysis to compare multiple datasets. Asterisks indicate statistical significance compared to the control group (** $P < 0.01$).

(alkali) (Pandey, Shrestha et al. 2023 [DOI](#)). In contrast, our survey revealed that five mutants (*Ir7g*¹, *Ir25a*², *Ir51b*¹, *Ir56d*¹, and *Ir76b*¹) exhibited strong defects in firing in response to cholesterol (**Figure 2A** [DOI](#)). Two of the mutations, *Ir25a*² and *Ir76b*¹, disrupt co-receptors that are necessary for sensing most attractive and aversive tastants (Ganguly, Pang et al. 2017 [DOI](#), Lee, Poudel et al. 2018 [DOI](#), Dhakal, . 2021, Shrestha and Lee 2021 [DOI](#), Stanley, Ghosh et al. 2021 [DOI](#), Aryal, Dhakal et al. 2022 [DOI](#), Xiao, Baik et al. 2022 [DOI](#), Li, Sun et al. 2023 [DOI](#), Pandey, Shrestha et al. 2023 [DOI](#), Pradhan, Shrestha et al. 2024 [DOI](#)). In further support of the roles of these five IRs for detecting cholesterol, we observed similar phenotypes resulting from mutation of additional alleles (*Ir7g*², *Ir51b*², *Ir56d*², and *Ir76b*²) (Zhang, Ni et al. 2013 [DOI](#), Sánchez-Alcañiz, Silbering et al. 2018 [DOI](#), Dhakal, Sang et al. 2021 [DOI](#), Pradhan, Shrestha et al. 2024 [DOI](#)), or due to placing the mutation (*Ir25a*²) *in trans* with a deficiency (Df) spanning the locus (**Figure 2B** [DOI](#)). Furthermore, using phytosterol stigmasterol to stimulate S6, S7, and S10 sensilla, we confirmed that the five mutants exhibited consistent phenotypes, underscoring the specificity of these IRs in sterol detection (**Figure 2—figure supplement 1C—E** [DOI](#)).

To provide additional verification that the phenotypes exhibited by the *Ir25a*², *Ir56d*¹, and *Ir76b*¹ mutants were attributed to the loss of *Ir25a*, *Ir56d*, and *Ir76b*, we conducted rescue experiments. To do so, we used *GAL4* lines specific to each gene to drive the respective wild-type *UAS-cDNAs* in the corresponding mutant backgrounds. We found that the responses to cholesterol were fully restored in S7 sensilla stimulated with 0.1% cholesterol (**Figures 2D and F** [DOI](#)).

To evaluate the dose-dependent defects exhibited by the mutants, we performed tip recordings to examine the neuronal responses of S7 sensilla to a spectrum of cholesterol percentages (10⁻⁵ to 10⁻¹). All five mutants exhibited significantly reduced neuronal firing in response to cholesterol percentages over a 100-fold range (10⁻³ to 10⁻¹; **Figure 2E** [DOI](#)). However, at lower percentages of cholesterol (10⁻⁵ and 10⁻⁴), all five IR mutants did not vary significantly from the control (**Figure 2E** [DOI](#)).

IRs required in B GRNs for cholesterol-induced neuronal firing

To test whether the IRs function in B GRNs, we used two approaches: RNA interference (RNAi) and gene rescue experiments. To knock down gene expression in B GRNs, we took advantage of the *Gr33a-GAL4* and found that targeting any of the five genes dramatically reduced action potentials in S7 sensilla in response to 0.1% cholesterol (**Figure 2C** [DOI](#)). In contrast, when we used a D GRN driver (*ppk23-GAL4*) in combination with the same *UAS-RNAi* lines, there was no decrease in neuronal firing. To perform gene rescue experiments, we used the *Gr33a-GAL4* to express each wild-type cDNA transgene in the corresponding mutant background and performed tip recordings. In all cases, we rescued the mutant phenotypes (**Figures 2D and F** [DOI](#)). Thus, we conclude that the IRs function in B GRNs.

Given that the five IRs are required in B GRNs, it stands to reason that they are expressed in these neurons. Indeed, *Ir7g*, *Ir25a*, *Ir51b*, and *Ir76b* have been shown previously to be expressed in B GRNs (Lee, Poudel et al. 2018 [DOI](#), Dhakal, Sang et al. 2021 [DOI](#), Pradhan, Shrestha et al. 2024 [DOI](#)). However, *Ir56d*, which has a role in sweet-sensing A GRNs, has not. To explore this possibility, we performed double-labeling experiments. We expressed *UAS-dsRed* under the control of the *Ir56d-GAL4* and did so in flies that included a B GRN reporter (*Gr66a-I-GFP*). Each of the two bilaterally symmetrical labella contain 11 S-type sensilla, 11 I-type sensilla and 9 L-type sensilla. We found that 10.7 ± 1.4 cells co-expressed both the dsRed and GFP markers (**Figure 2—figure supplement 1F** [DOI](#)). By tracing dendrites from individual GFP-expressing cells, we identified the specific sensilla innervated by each marker. Most S2, S3, S4, S6, and S7 sensilla that expressed the *Ir56d* reporter were co-labeled with the B GRN reporter. Thus, the B GRNs in the two sensilla that elicited the highest frequency of cholesterol-induced action potentials (S6 and S7) were labeled by the *Ir56d* reporter.

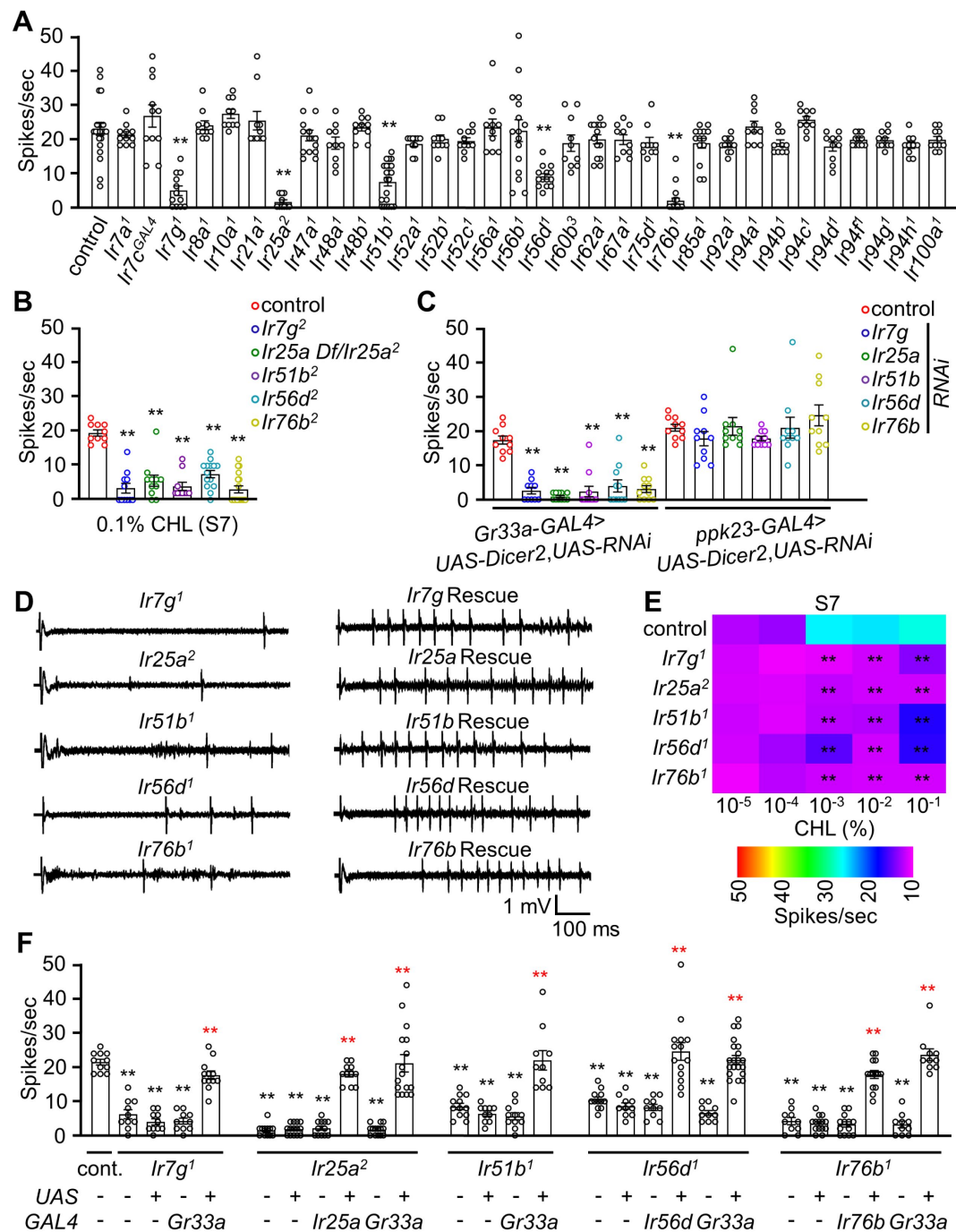


Figure 2.

Ionomotropic receptors (IRs) are responsible for sensing cholesterol.

(A) Tip recording analysis of 0.1% cholesterol with control and 32 *Ir* mutants on S7 sensilla ($n = 10-16$). (B) Tip recording analysis of the *Ir7g²*, *Ir25a Df/Ir25a²*, *Ir51b²*, *Ir56d²*, and *Ir76b²* ($n = 10-16$). (C) Electrophysiology of the RNAi lines of the *Ir7g*, *Ir25a*, *Ir51b*, *Ir56d*, and *Ir76b* driven by the *Gr33a-GAL4* and *ppk23-GAL4* in S7 sensilla ($n = 10-16$). (D) Representative sample traces of (F) for control, mutants, and rescue lines. (E) Heatmap representing the dose responses of the control and the candidate mutants *Ir7g¹*, *Ir25a²*, *Ir51b¹*, *Ir56d¹*, and *Ir76b¹* analyzed using electrophysiology ($n = 10-16$). (F) Tip recordings of control, *Ir7g¹*, *Ir25a²*, *Ir51b¹*, *Ir56d¹*, *Ir76b¹*, and genetically recovered flies driven by their own *GAL4* and *Gr33a-GAL4* on S7 sensilla with 0.1% cholesterol ($n = 10-14$). All error bars represent SEM. Single-factor ANOVA was combined with Scheffe's post hoc analysis to compare multiple datasets. Asterisks indicate statistical significance compared to the control group; the red asterisks indicate statistical significance between the control and the rescued flies ($**P < 0.01$).

IRs required for avoiding the taste of cholesterol

The requirement for the five IRs for cholesterol-induced action potentials in B GRNs suggests that cholesterol is an aversive taste. To explore this question, we used the well-established binary choice assay in which we allowed flies to choose between 2 mM sucrose alone or 2 mM sucrose mixed with various percentages of cholesterol. We mixed the two food alternatives with either blue or red food dye so we could inspect the flies' abdomens to assess which option they consumed (Aryal, Dhakal et al. 2022 [DOI](#)). At the lowest percentage tested ($10^{-5}\%$), flies showed only a slight aversion to cholesterol-containing food (**Figure 3A** [DOI](#)). As cholesterol concentration increased, they showed a dose-dependent aversion, with a very strong aversion at 0.1% ($PI = -0.72 \pm 0.03$; **Figure 3A** [DOI](#)). Both male and female flies showed comparable avoidance responses to 0.1% cholesterol, indicating the behavior is not sex-specific (**Figure 3B** [DOI](#)). The aversion was not due to the M β CD used to dissolve the cholesterol since the flies were indifferent to sucrose alone versus sucrose plus M β CD (**Figure 3—figure supplement 1A** [DOI](#)). Moreover, the flies showed similar levels of aversion to sucrose plus cholesterol versus either sucrose alone (**Figure 3A** [DOI](#)) or sucrose plus M β CD (**Figure 3—figure supplement 1B** [DOI](#)). The dye used in the study also did not alter the behavioral response (**Figure 3—figure supplement 1C** [DOI](#)).

To determine the impact of inhibiting B GRNs on gustatory behavior, we used the *Gr33a-GAL4* to drive an expression of *UAS-Kir2.1*. As a control, we also inactivated other classes of GRNs and found that expression of *kir2.1* in A GRNs (*Gr64f-GAL4*), C GRNs (*ppk23-GAL4*), and D GRNs (*ppk28-GAL4*) had no impact on cholesterol avoidance (**Figure 3C** [DOI](#)). Surprisingly, inhibiting B GRNs (*Gr33a-GAL4*) not only eliminated cholesterol avoidance, it caused the flies to exhibit a preference for cholesterol-containing food, thereby unmasking some unknown attractive mechanism.

We also set out to assess the requirements for the five IRs for cholesterol taste. Therefore, we performed two-way choice assays. All mutants showed defects in avoiding the sucrose containing 0.1% cholesterol over sucrose alone (**Figures 3D and E** [DOI](#)). We then tested the behavior of the mutants across a range of cholesterol percentages ($10^{-5}\%$ to $10^{-1}\%$). We found that the *Ir25a*², *Ir51b*¹, *Ir56d*¹, and *Ir76b*¹ mutants exhibited reduced aversion across all cholesterol concentrations tested (**Figure 3F** [DOI](#)). However, *Ir7g*¹ showed a deficit in behavioral avoidance only at higher cholesterol percentages (10^{-2} and $10^{-1}\%$; **Figure 3F** [DOI](#)). To test for rescue, we expressed the wild-type cDNA of each IR using its respective *GAL4*. The avoidance deficiencies in the five IR mutants were fully restored (**Figure 3G** [DOI](#)).

To determine whether the IRs function in B GRNs, we performed rescue experiments and RNAi. In all cases, the mutant phenotypes were rescued using the B GRN driver (*Gr33a-GAL4*) in combination with the corresponding *UAS-cDNA* (**Figure 3G** [DOI](#)). To perform RNAi, we also took advantage of the *Gr33a-GAL4*. We found that knockdown of each *Ir* in B GRNs eliminated cholesterol avoidance (**Figure 3—figure supplement 1D** [DOI](#)). However, RNAi knockdown in D GRNs (*ppk23-GAL4*) had no impact on the repulsion to cholesterol (**Figure 3—figure supplement 1E** [DOI](#)). Thus, we conclude that the *Irs* function in B GRNs.

We also conducted binary food choice assays to address whether olfaction contributed to the avoidance of cholesterol. We found that the *orco* null mutant (*orco*¹), which disrupts the olfactory co-receptor (ORCO) broadly required for olfaction, exhibited cholesterol repulsion similar to control flies (**Figure 3—figure supplement 1F** [DOI](#)). Consistent with these findings, surgically ablating the antennae and maxillary palps, which are the main olfactory organs, did not diminish cholesterol avoidance (**Figure 3—figure supplement 1G** [DOI](#)).

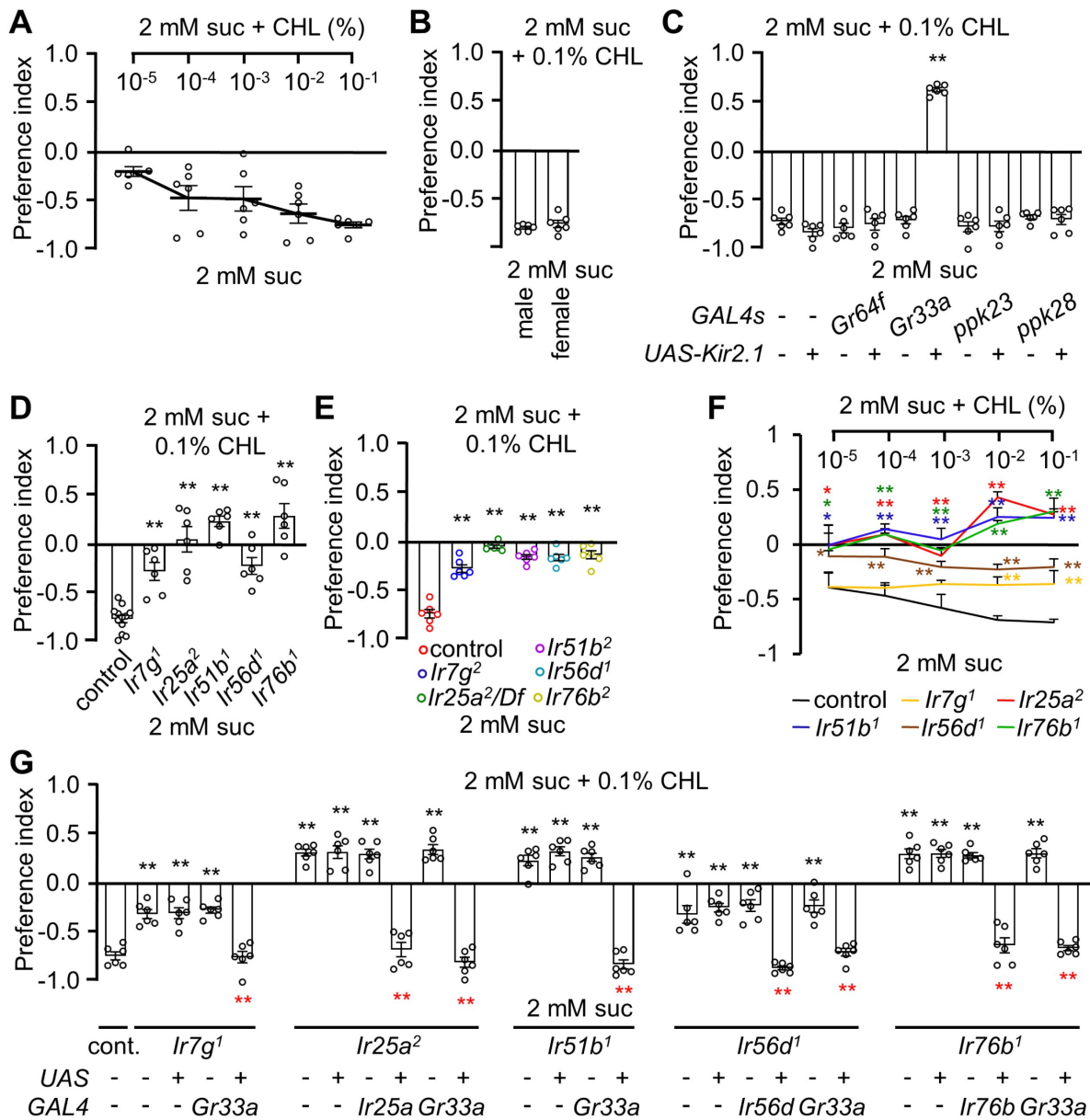


Figure 3.

***Ir7g*, *Ir25a*, *Ir51b*, *Ir56d*, and *Ir76b* are required for the perception of cholesterol.**

(A) Behavioral analysis of the *w¹¹¹⁸* adult flies toward different doses of cholesterol. Sucrose (2 mM) was included on both sides (n = 6). (B) Sex-wise feeding assay analysis toward 0.1% cholesterol (n = 6). (C) Binary food choice assay of specific gustatory receptor neuron (GRN)-ablated flies toward 0.1% cholesterol; +/- indicates the presence or absence of the transgene, respectively (n = 6). (D) Two-way choice assay of the control, *Ir7g¹*, *Ir25a²*, *Ir51b¹*, *Ir56d¹*, and *Ir76b¹* toward 0.1% cholesterol (n = 6). (E) Feeding assay analysis of *Ir7g²*, *Ir25a²/Df*, *Ir51b²*, *Ir56d²*, and *Ir76b²* (n = 6). (F) Dose response of control, *Ir7g¹*, *Ir25a²*, *Ir51b¹*, *Ir56d¹*, and *Ir76b¹* toward different concentrations of cholesterol (10⁻⁵%, 10⁻⁴%, 10⁻³%, 10⁻²%, and 10⁻¹%) via binary food choice assay (n = 6). (G) Rescue of *Ir7g¹*, *Ir25a²*, *Ir51b¹*, *Ir56d¹*, and *Ir76b¹* defects by expressing the wild-type cDNA under the control of the respective GAL4 drivers (n = 6). All error bars represent SEM. Single-factor ANOVA was combined with Scheffe's post hoc analysis to compare multiple datasets. Asterisks indicate statistical significance compared to the control group; the red asterisks indicate statistical significance between the control and the rescued flies (***P* < 0.01).

Ectopic co-expression of two sets of IRs confers responses to cholesterol

To explore whether *Ir7g*, *Ir25a*, *Ir51b*, *Ir56d*, and *Ir76b* are sufficient to confer cholesterol sensitivity to GRNs that are normally unresponsive to cholesterol, we conducted ectopic expression experiments. We expressed the five *Irs* in all B GRNs (*Gr33a-GAL4*; [Figure 4A](#)) or in all A GRNs (*Gr5a-GAL4*; [Figure 4C](#)) and then characterized cholesterol-induced action potentials by focusing on cholesterol-insensitive I-type sensilla. Introducing all five *Irs* into cholesterol-insensitive, I9 sensilla elicited strong responses ([Figure 4B](#)). Misexpression of just *Ir7g*, *Ir51b*, and *Ir56d* also replicated cholesterol-induced responses ([Figure 4B](#)), presumably because *Ir25a* and *Ir76b* are endogenously expressed in GRNs in these sensilla (Lee, Poudel et al. 2018). Expression of any one of these *Irs* (*Ir7g*, *Ir51b*, and *Ir56d*) or combining *Ir7g* and *Ir51b* was insufficient to induce cholesterol sensitivity ([Figure 4B](#)). Of significance, we found that combining *Ir56d* with either *Ir51b* or *Ir7g* conferred cholesterol sensitivity to I9 sensilla ([Figure 4B](#)).

L-type sensilla are missing B GRNs and are unresponsive to cholesterol. Therefore, we misexpressed all five *Irs* in A GRNs using the *Gr5a-GAL4* ([Figure 4C](#)) and characterized action potentials in L6 sensilla. Ectopic expression of the *Irs* in the A GRNs also bestowed responsiveness to cholesterol ([Figure 4D](#)). Consistent with the results in B GRNs, co-expression of *Ir51b* and *Ir56d*, or *Ir7g* and *Ir56d*, was sufficient to confer cholesterol sensitivity ([Figure 4D](#)). This indicates that either of two groups of four IRs (*Ir7g*, *Ir25a*, *Ir56d*, *Ir76b* or *Ir25a*, *Ir51b*, *Ir56d*, *Ir76b*) is sufficient to comprise a functional cholesterol receptor.

Inducing attraction to cholesterol

Activation of A GRNs by sugars and several other attractive chemicals promotes feeding. Given that ectopic expression of *Ir7g* and *Ir56d*, or *Ir51b* and *Ir56d* alone was sufficient to induce a response to cholesterol in A GRNs, we investigated whether this would elicit attraction toward cholesterol. Control flies exhibited a preference for 2 mM sucrose alone over 2 mM sucrose laced with cholesterol ([Figure 4E](#)). *Ir56d*¹ and *Ir7g*¹ mutant flies showed a slight avoidance of cholesterol-laced food. Flies carrying the *Ir56d*¹ or the *Ir7g*¹ mutation and expressing both *UAS-Ir7g* and *UAS-Ir51b* in A GRNs exhibited similar behavior as *Ir56d*¹ or *Ir7g*¹ flies. However, when we introduced *Ir7g* and *Ir56d*, or *Ir51b* and *Ir56d* in the mutants, the flies exhibited attraction to the cholesterol-laced food ([Figure 4E](#)).

Discussion

The impact of cholesterol on an animal's health depends on the concentration of cholesterol that is consumed (Huang, Hsu et al. 2024). While low levels are crucial, animals must avoid consuming excessive cholesterol (Beynen 1988, Soliman 2018, Zhang, Yuan et al. 2019). This differential effect of cholesterol is reminiscent of the impact of Na⁺ and Ca²⁺ on health, depending on their concentration (Zhang, Ni et al. 2013, Lee, Poudel et al. 2018). Flies have a bivalent reaction to Na⁺ depending on concentration but only avoid high Ca²⁺ and are indifferent to low Ca²⁺. Therefore, it was an open question as to whether flies have the capacity to taste cholesterol, and if so, whether they are endowed with the capacity to respond differentially to low and high concentrations or only avoid high cholesterol.

Several biochemical and biophysical studies focusing on the mammalian taste receptors T2R4 and T2R14 demonstrate that these receptors can bind and be activated or modulated by cholesterol (Pydi, Jafurulla et al. 2016, Shaik, Jaggupilli et al. 2019, Kim, Gumpfer et al. 2024). While T2R4 and T2R14 are expressed in the taste system, they are expressed at high levels extraorally,

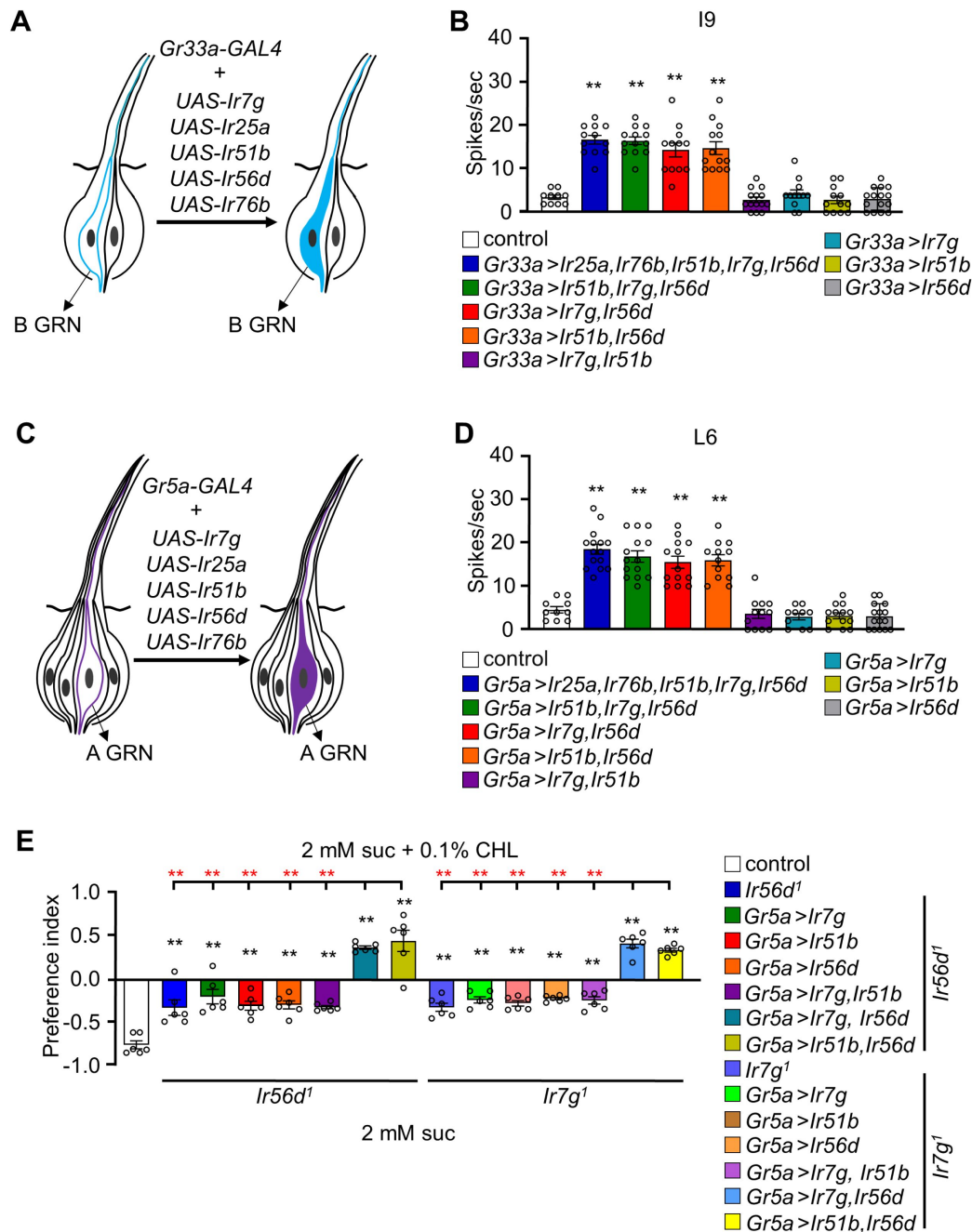


Figure 4.

Recapitulation of *Ir7g*, *Ir25a*, *Ir51b*, *Ir56d*, and *Ir76b* on L- and I-type sensilla.

(A) Schematic representation of heteromultimeric association of *Ir7g*, *Ir51b*, and *Ir56d* in I-type sensilla for cholesterol taste processing using *Gr33a-GAL4*. (B) Tip recordings were conducted from I9 sensilla after activation with 0.1% cholesterol by overexpression of *UAS-Ir7g*, *UAS-Ir25a*, *UAS-Ir51b*, *UAS-Ir56d*, and *UAS-Ir76b* in bitter-sensing gustatory receptor neurons (GRNs) via *Gr33a-GAL4* ($n = 10-16$). (C) Schematic presentation of misexpression of *Ir7g*, *Ir51b*, and *Ir56d* in L-type sensilla using *Gr5a-GAL4*. (D) Tip recording of indicated flies from L6 sensilla *Gr5a-GAL4* ($n = 10-16$). (E) Behavioral ectopic analysis for attraction or aversion to 0.1% cholesterol in flies misexpressing *Ir7g*, *Ir51b*, and *Ir56d* in sweet-sensing GRNs (*Gr5a-GAL4*). The IRs were ectopically expressed in an *Ir56d*¹ and *Ir7g*¹ mutant background ($n = 6$). The red asterisks indicate the comparison of the combination of two *UAS* lines (*Ir7g*, *Ir56d* and *Ir51b*, *Ir56d*) driven by *Gr5a-GAL4* with all the single *UAS* line including combination of *Ir7g* and *Ir51b*. All error bars represent the SEM. Single-factor ANOVA was combined with Scheffe's post hoc analysis to compare multiple datasets. Asterisks indicate statistical significance compared with the control (** $P < 0.01$).

such as in airway epithelial cells, pulmonary artery smooth muscle cells, and breast epithelial cells (Hariri, McMahon et al. 2017 [↗](#), Jaggupilli, Singh et al. 2017 [↗](#), Singh, Shaik et al. 2020 [↗](#)). Therefore, it has been thought that these receptors may function in interoception, enabling the body to sense and respond to internal levels of cholesterol. Currently, evidence that these or other receptors function in cholesterol taste is lacking in mammals or any other animal.

Our research highlights the discovery that flies reject higher levels of cholesterol, but do not show attraction to low cholesterol even though flies cannot synthesize cholesterol and therefore must meet their needs for cholesterol from their diet. Nevertheless, the repulsion to a required substance is reminiscent of the fly's response to Ca^{2+} , which is aversive even though Ca^{2+} is required for life (Lee, Poudel et al. 2018 [↗](#)). Moreover, we discovered that the repulsion to high cholesterol is mediated through the taste system since cholesterol stimulates action potentials in a subset of GRNs. The class of GRN that is activated by cholesterol is the B class, which also responds to bitter chemicals and other aversive tastants. However, cholesterol stimulates only a subset of the bitter responsive GRNs.

An unexpected observation is that the inhibition of B GRNs with Kir2.1 not only eliminates cholesterol repulsion but causes cholesterol to become highly attractive. This unmasking of an attractive mechanism for cholesterol when B GRNs are inhibited raises questions about the underlying neural circuitry and molecular mechanisms. Notably, this effect is unlikely to be due to cholesterol-induced activation of GRNs that promote feeding, since cholesterol does not activate any GRN in L-type sensilla, which are devoid of B GRNs, but include three types of attractive GRNs: A GRNs, C GRNs and other class of GRNs (E) (Montell 2021 [↗](#)). Thus, we suggest that the attraction to cholesterol, which is unmasked by inhibition of B GRNs, occurs through a mechanism postsynaptic to the GRNs. Understanding the attractive mechanisms could provide valuable insights into how *Drosophila* regulates cholesterol intake based on internal nutritional states. This is particularly relevant given that *Drosophila*, like other insects, are cholesterol auxotrophs and must obtain sterols from their diet (Carvalho, Schwudke et al. 2010 [↗](#)). Furthermore, elucidating the neural and molecular basis of cholesterol attraction and its potential modulation by internal metabolic states in *Drosophila* has the potential to reveal evolutionarily conserved mechanisms.

A key issue concerns the molecular identity of the *Drosophila* cholesterol taste receptor. We addressed this question using both electrophysiological and behavioral approaches to assess the impact of mutating genes encoding receptors belonging to the major families of fly taste receptors. We found that five IRs are required for cholesterol taste. These include two broadly required co-receptors (IR25a and IR76b) and three other receptors (IR7g, IR51b and IR56d). The contribution of five IRs to cholesterol taste was unanticipated since IRs are thought to be tetramers (Wicher and Miazzi 2021 [↗](#)). Therefore, we conducted a series of ectopic expression experiments to determine whether all five IRs were necessary to confer cholesterol sensitivity to GRNs that do not normally respond to cholesterol. We found that either of two combinations of four IRs was sufficient to endow cholesterol responsiveness to GRNs. Three of the IRs were common to both receptors (IR25a, IR56d, IR76b). However, addition of either IR7g or IR51b as the fourth IR was necessary to generate a cholesterol receptor. It remains to be determined as to why there are two cholesterol receptors, why both are required for cholesterol taste, and why four rather than three IR subunits comprise each receptor.

The findings reported here raise the question as to whether mammals such as mice and humans perceive cholesterol through the sense of taste. It is notable that in flies cholesterol taste depends on B GRNs, which also sense bitter compounds, and that in mammals T2Rs are activated by cholesterol, since this family of receptors also respond to bitter compounds. Therefore, it is intriguing to speculate that cholesterol taste may be aversive in humans.

Materials and Methods

REAGENT or RESOURCE	SOURCE or REFERENCE	IDENTIFIER
Antibodies		
Rabbit anti-DsRed	TaKaRa	Cat #632496; RRID: AB_10013483
Goat anti-rabbit Alexa Fluor 568	Invitrogen (Thermo Fisher)	A11011
Mouse anti-GFP	Molecular probes	Cat # A11120; RPID: AB_221568
Goat anti-mouse Alexa Fluor 488	Invitrogen (Thermo Fisher)	A-11029
Chemicals, peptides, and recombinant proteins		
Cholesterol	Sigma-Aldrich Co.	Cat# C4951
Sucrose	Sigma-Aldrich Co.	Cat# S9378
Tricholine citrate	Sigma-Aldrich Co.	Cat# T0252
Stigmasterol	Sigma-Aldrich Co.	Cat# S2424
Sulforhodamine B	Sigma-Aldrich Co.	Cat# 230162
Brilliant blue FCF	Wako Pure Chemical Industry Ltd.	Cat# 027-12842
Methyl beta cyclodextrin	Sigma-Aldrich Co.	Cat# 332615
Paraformaldehyde	Electron Microscopy Sciences	Cat # 15710
Goat Serum, New Zealand origin	Gibco	Cat # 16210064
Experimental models: Organisms/strains		
Genotype	Source	Identifier
<i>Ir7a¹</i>	Dr. Y. Lee (Rimal, Sang et al. 2019)	
<i>Ir7g¹: y¹ w[*] Mi{y⁺mDint2}=MIC}Ir7g^{M106687}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:42420

<i>Ir8a</i> ¹ :w[*]Tl{w[+m*]=Tl}Ir8a[1];Bl[1]L[2]/CyO	Bloomington <i>Drosophila</i> Center Stock	BDSC:23842
<i>Ir10a</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir10a ^{MB03273}	Bloomington <i>Drosophila</i> Center Stock	BDSC:41744
<i>Ir21a</i> ¹ :w ¹¹¹⁸ ;PBac{w ^{+mC} =PB}Ir21a ⁶⁰²⁷²⁰	Bloomington <i>Drosophila</i> Center Stock	BDSC:10975
<i>Ir25a</i> ²	Dr. L. Voshall (Benton, Vannice et al. 2009)	
<i>Ir47a</i> ¹	Dr. Y. Lee (Rimal, Sang et al. 2019)	
<i>Ir48a</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir48a ^{MB09217}	Bloomington <i>Drosophila</i> Center Stock	BDSC:26453
<i>Ir48b</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir48b ^{MB02315}	Bloomington <i>Drosophila</i> Center Stock	BDSC:23473
<i>Ir51b</i> ¹ :w ¹¹¹⁸ ;PBac{w ^{+mC} =PB}row ⁶⁰⁰³⁸⁷ Ir51b ⁶⁰⁰³⁸⁷	Bloomington <i>Drosophila</i> Center Stock	BDSC:10046
<i>Ir52a</i> ¹	Dr. Y. Lee (Rimal and Lee 2019)	
<i>Ir52b</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir52b ^{MB02231} /SM6a	Bloomington <i>Drosophila</i> Center Stock	BDSC:25212
<i>Ir52c</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir52c ^{MB04402}	Bloomington <i>Drosophila</i> Center Stock	BDSC:24580
<i>Ir56a</i> ¹	Dr. Y. Lee (Rimal and Lee 2019)	
<i>Ir56b</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir56b ^{MB09950}	Bloomington <i>Drosophila</i> Center Stock	BDSC:27818
<i>Ir56d</i> ¹ :w[*];Ir56d[1]	Bloomington <i>Drosophila</i> Center Stock	BDSC:81249
<i>Ir60b</i> ³	Dr. Y. Lee (Sang, Dhakal et al. 2024)	

<i>Ir62a</i> ¹ :y ¹ w [*] ;Mi{y ^{+mDint2} =MIC}Ir62a ^{MI00895} Im1 ^{MI00895} /TM3, Sb ¹ Ser ¹	Bloomington <i>Drosophila</i> Center Stock	BDSC:32713
<i>Ir67a</i> ¹ :y ¹ w [*] ;Mi{y ^{+mDint2} =MIC}Ir67a ^{MI11288}	Bloomington <i>Drosophila</i> Center Stock	BDSC:56583
<i>Ir75d</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir75d ^{MB04616}	Bloomington <i>Drosophila</i> Center Stock	BDSC:24205
<i>Ir76b</i> ¹	Dr. C. Montell (Zhang, Ni et al. 2013)	
<i>Ir85a</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir85a ^{MB04613} Pif1A ^{MB04613}	Bloomington <i>Drosophila</i> Center Stock	BDSC:24590
<i>Ir92a</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir92a ^{MB03705}	Bloomington <i>Drosophila</i> Center Stock	BDSC:23638
<i>Ir94a</i> ¹	Dr. Y. Lee (Rimal and Lee 2019)	
<i>Ir94b</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir94b ^{MB02190}	Bloomington <i>Drosophila</i> Center Stock	BDSC:23424
<i>Ir94c</i> ¹	Dr. Y. Lee (Rimal and Lee 2019)	
<i>Ir94d</i> ¹ :y ¹ w ¹ ;Mi{y ^{+mDint2} =MIC}Ir94d ^{MI01659} CG17380 ^{MI01659}	Bloomington <i>Drosophila</i> Center Stock	BDSC:33132
<i>Ir94f</i> ¹ :y ¹ w [*] ;Mi{y ^{+mDint2} =MIC}Ir94f ^{MI00928}	Bloomington <i>Drosophila</i> Center Stock	BDSC:33095
<i>Ir94g</i> ¹ :w ¹¹¹⁸ ;Mi{GFP ^{E.3xP3} =ET1}Ir94g ^{MB07445}	Bloomington <i>Drosophila</i> Center Stock	BDSC:25551
<i>Ir94h</i> ¹	Dr. Y. Lee (Rimal and Lee 2019)	
<i>Ir100a</i> ¹ :w ¹¹¹⁸ ;P{w ^{+mC} =EP}Ir100a ^{G19846} CG42233 ^{G19846}	Bloomington <i>Drosophila</i> Center Stock	BDSC:31853
<i>UAS-Ir25a</i>	Dr. Y. Lee (Lee, Poudel et al. 2018)	

<i>UAS-Ir51b</i>	Dr. Y. Lee (Dhakal, Sang et al. 2021)	
<i>Gr33a¹</i>	Dr. C. Montell (Moon, Lee et al. 2009)	
<i>Gr33a-GAL4</i>	Dr. C. Montell (Moon, Lee et al. 2009)	
<i>Gr47a¹</i>	Dr. C. Montell (Lee, Moon et al. 2015)	
<i>elav-GAL4;UAS-Dicer2</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:25750
<i>Gr39a¹</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:10562
<i>Gr93a³</i>	Dr. Y. Lee (Lee, Moon et al. 2009)	
<i>UAS-Kir2.1</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:6596
Δ <i>Gr32a</i>	Dr. H. Amrein (Miyamoto and Amrein 2008)	
<i>Gr66a^{ex83}</i>	Dr. C. Montell (Moon, Köttgen et al. 2006)	
<i>Gr89a¹</i>	Korea <i>Drosophila</i> Resource Center (KDRC)	KDRC: (Sung, Jeong et al. 2017)
<i>Ir7c^{GAL4}</i>	Dr. M. Gordon (McDowell, Stanley et al. 2022)	
<i>Ir20a¹</i>	Dr. A. Dahanukar (Ganguly, Pang et al. 2017)	
<i>Ir25a-GAL4</i>	Dr. L. Voss hall (Benton, Vannice et al. 2009)	

<i>UAS-Ir76b</i>	Dr. C. Montell (Moon, Köttgen et al. 2006)	
<i>Ir76b-GAL4</i>	Dr. C. Montell (Moon, Köttgen et al. 2006)	
<i>ppk23-GAL4</i>	Dr. K. Scott (Thistle, Cameron et al. 2012)	
<i>ppk28-GAL4</i>	Dr. H. Amrein (Cameron, Hiroi et al. 2010)	
<i>Gr5a-GAL4</i>	Dr. H. Amrein (Dahanukar, Foster et al. 2001)	
<i>UAS-Kir2.1</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:6595
<i>Ir7g²</i>	Dr. Y. Lee (Pradhan, Shrestha et al. 2024)	
<i>UAS-Ir7g</i>	Dr. Y. Lee (Pradhan, Shrestha et al. 2024)	
<i>UAS-Ir56d</i>	Dr. R. Benton (Sánchez-Alcañiz, Silbering et al. 2018)	
<i>Ir56d-GAL4</i>	Korea <i>Drosophila</i> Resource Center	KDRC:2307
<i>Ir56d²</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:81250
<i>Ir51b²</i>	Dr. Y. Lee (Dhakal, Sang et al. 2021)	
<i>BC/Cyo;Gr66a-I-GFP,UAS-dsred/TM6b</i>	Dr. J.R. Carlson (Weiss, Dahanukar et al. 2011)	
<i>Ir7g RNAi</i>	Vienna <i>Drosophila</i> Resource Center	VDRC:100885
<i>Ir25a RNAi</i>	Vienna <i>Drosophila</i> Resource Center	VDRC:106731

<i>Ir51b RNAi</i>	Vienna <i>Drosophila</i> Resource Center	VDRC:29984
<i>Ir56d RNAi</i>	Vienna <i>Drosophila</i> Resource Center	VDRC6112
<i>Ir76b RNAi</i>	Vienna <i>Drosophila</i> Resource Center	VDRC8433
<i>trpA1</i> ¹	N/A (Kwon, Shim et al. 2008)	
<i>trpI</i> ²⁹¹³⁴	N/A (Niemeyer, Suzuki et al. 1996)	
<i>trpy</i> ¹	N/A (Akitake, Ren et al. 2015)	
<i>amo</i> ¹	N/A (Watnick, Jin et al. 2003)	
<i>iav</i> ³⁶²¹	N/A (Gong, Son et al. 2004)	
<i>nan</i> ^{36a}	N/A (Kim, Chung et al. 2003)	
<i>trp</i> ³⁴³	N/A (Tracey Jr, Wilson et al. 2003)	
<i>pyx</i> ³	N/A (Lee, Lee et al. 2005)	
<i>wtrw</i> ^{BX}	N/A (Kim, Lee et al. 2010)	
<i>pain</i> ²	N/A (Tracey Jr, Wilson et al. 2003)	
Software and algorithms		
<i>Origin Pro Version</i>	Dr. Y. Lee	https://www.originlab.com (Origin Lab Corporation, RRID: SCR 002815)
<i>GraphPad Prism Version 8.0</i>	Dr. Y. Lee	https://www.graphpad.com (RRID: SCR 002798)
Autospike 3.1 software	Dr. Y. Lee	https://www.syntech.co.za/

Key resources table

Chemical reagents

The following chemicals and reagents were purchased from Sigma-Aldrich: cholesterol (catalog no. C4951), M β CD (catalog no. 332615), stigmasterol (catalog no. S2424), propionic acid (catalog no. 402907), butyric acid (catalog no. B103500), acetic acid (catalog no. A8976), caffeine (catalog no. C0750), denatonium benzoate (catalog no. D5765), sulforhodamine B (catalog no. 230162), tricholine citrate (TCC; catalog no. T0252), umbelliferone (catalog no. 24003), berberine sulfate hydrate (catalog no. B0451), cholesteroloquinone diphosphate salt (catalog no. C6628), lobeline hydrochloride (catalog no. 141879), quinine hydrochloride dihydrate (catalog no. Q1125), papaverine hydrochloride (catalog no. P3510), strychnine hydrochloride (catalog no. S8753), coumarin (catalog no. C4261), and sucrose (catalog no. S9378). Brilliant Blue FCF (catalog no. 027-12842) was purchased from Wako Pure Chemical Industry Ltd. The following antibodies were purchased from respective sources: mouse anti-GFP antibody (Molecular Probes, catalog no. A11120), rabbit anti-DsRed (TaKaRa Bio, catalog no. 632496), goat anti-mouse Alexa Fluor 488 (Thermo Fisher Scientific, catalog no. A11029), and goat anti-rabbit Alexa Fluor 568 (catalog no. A11011, Thermo Fisher Scientific/Invitrogen).

Binary food choice assay

In accordance with a previous study, we carried out experiments involving binary food choice tests (Aryal, Dhakal et al. 2022 [\[1\]](#)). To initiate the experiment, a group of 50–70 flies (aged 3–6 days, consisting of both males and females) were subjected to an 18-h period of fasting in a controlled humidity chamber. The subsequent procedures included the preparation of two distinct food sources, both incorporating 1% agarose as the base. The first food source was enriched with 2 mM sucrose, and the second source contained different concentrations of cholesterol in addition to 2 mM sucrose. To distinguish between these two food sources, we introduced blue food coloring dye (0.125 mg/mL brilliant blue FCF) to one and red food coloring dye (0.1 mg/mL sulforhodamine B) to the other. We evenly distributed these prepared solutions into the wells of a 72-well microtiter dish (Thermo Fisher Scientific, catalog no. 438733), alternating between the two options. Approximately 50–70 starved flies were introduced to the plate within approximately 30 min of food preparation. The flies were allowed to feed at room temperature (25°C) for 90 min, which occurred in a dark, humid environment to maintain consistent conditions. Afterward, the tested flies were carefully frozen at –20°C for further analysis. With the aid of a stereomicroscope, we observed and categorized the coloration of their abdomens as either blue (N_B), red (N_R), or purple (N_P). For each fly, we calculated the PI, a value derived from the combinations of dye and tastant, as follows: $(N_B - N_R) / (N_R + N_B + N_P)$ or $(N_R - N_B) / (N_R + N_B + N_P)$. A PI of either 1.0 or –1.0 indicated a significant preference toward one of the food alternatives, and a PI of 0.0 signified no bias among the flies toward either option.

Tip recording assay

The electrophysiological procedure, specifically the tip recording assay, was conducted in strict adherence to previously established protocols (Moon, Köttgen et al. 2006 [\[2\]](#), Shrestha, Nhuchhen Pradhan et al. 2022 [\[3\]](#)). Flies of both sexes, aged between 4 and 7 days, were gently anesthetized on a bed of ice to facilitate the procedure. A reference glass electrode containing Ringer’s solution was meticulously inserted into the thoracic region of the flies. Subsequently, the electrode was incrementally advanced toward the proboscis of each fly. This precise process was repeated over multiple days to ensure the reliability and consistency of the results. To stimulate the sensilla, a recording pipette with a tip diameter ranging from 10 to 20 μ m was connected to a preamplifier. The pipette was filled with a blend of chemical stimulants dissolved in a 30 mM TCC solution, which served as the electrolyte solution. Signal amplification was achieved through a Syntech signal connection interface box and a band-pass filter spanning a range of 100–3000 Hz. These amplified signals were recorded at a sampling rate of 12 kHz and subsequently analyzed using

AutoSpike 3.1 software (Syntech). To ensure the integrity of the recorded signals, all recordings were carried out at regular 1-min intervals. This approach was meticulously enforced to guarantee the acquisition of precise and dependable data throughout the experiments.

Immunohistochemistry

Immunohistochemistry analysis was executed following established procedures (Lee, Kang et al. 2012 [↗](#)). The labellum or brain of the flies were meticulously dissected and fixed using a 4% paraformaldehyde solution (Electron Microscopy Sciences, catalog no. 15710) in PBS-T (1X phosphate-buffered saline containing 0.2% Triton X-100) for 25 min at 4°C. The fixed tissues were thoroughly rinsed three times with PBS-T for 15 min each, precisely bisected using a razor blade, and then incubated for 30 min at room temperature in a blocking buffer composed of 0.5% goat serum in 1X PBS-T. Primary antibodies (1:1000 dilution; mouse anti-GFP [Molecular Probes, catalog no. A11120] and rabbit anti-DsRed [TaKaRa Bio, catalog no. 632496]) were added to freshly prepared blocking buffer and incubated with the samples overnight at 4°C. After overnight incubation, the samples underwent an additional round of thorough washing with PBS-T at 4°C before being exposed to secondary antibodies (1:200 dilution in blocking buffer; goat anti-mouse Alexa Fluor 488 [Thermo Fisher Scientific, catalog no. A11029] and goat anti-rabbit Alexa Fluor 568 [Thermo Fisher Scientific/Invitrogen, catalog no. A11011]) for 4 h at 4°C. After another three rounds of washing with PBS-T, the tissues were immersed in 1.25X PDA mounting buffer (37.5% glycerol, 187.5 mM NaCl, and 62.5 mM Tris, pH 8.8) and examined using an inverted Leica LASX confocal microscope for visualization and analysis.

Quantification and statistical analyses

We processed and conducted data analysis using GraphPad Prism version 8.0 (RRID: SCR 002798). Each experiment was independently replicated on different days, and the number of trials for each experiment is indicated as data points on the graphs. Error bars on the graphs represent the standard error of the mean (SEM). Single-factor ANOVA was combined with Scheffe's post hoc analysis to compare multiple datasets. All statistical analyses were carried out using Origin (Origin Lab Corporation, RRID: SCR 002815). In the figures, asterisks are used to indicate statistical significance, with denotations of $*P < 0.05$ and $**P < 0.01$.

Additional files

Data availability

Source data for all figures contained in the manuscript have been deposited in 'figshare' (<https://doi.org/10.6084/m9.figshare.28293062> [↗](#)).

Supplementary figures

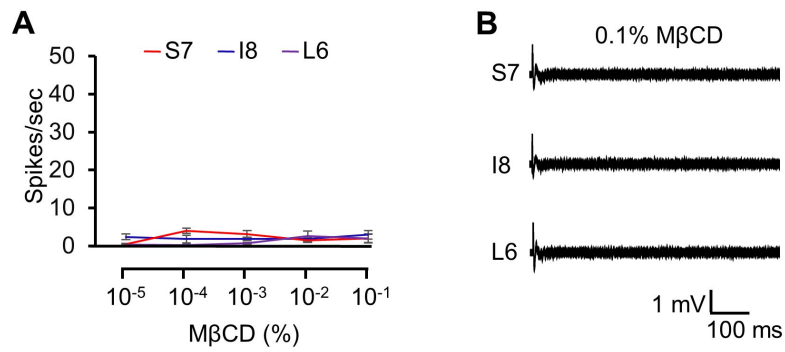


Figure 1—figure supplement 1.

Electrophysiological analysis of different doses of MβCD.

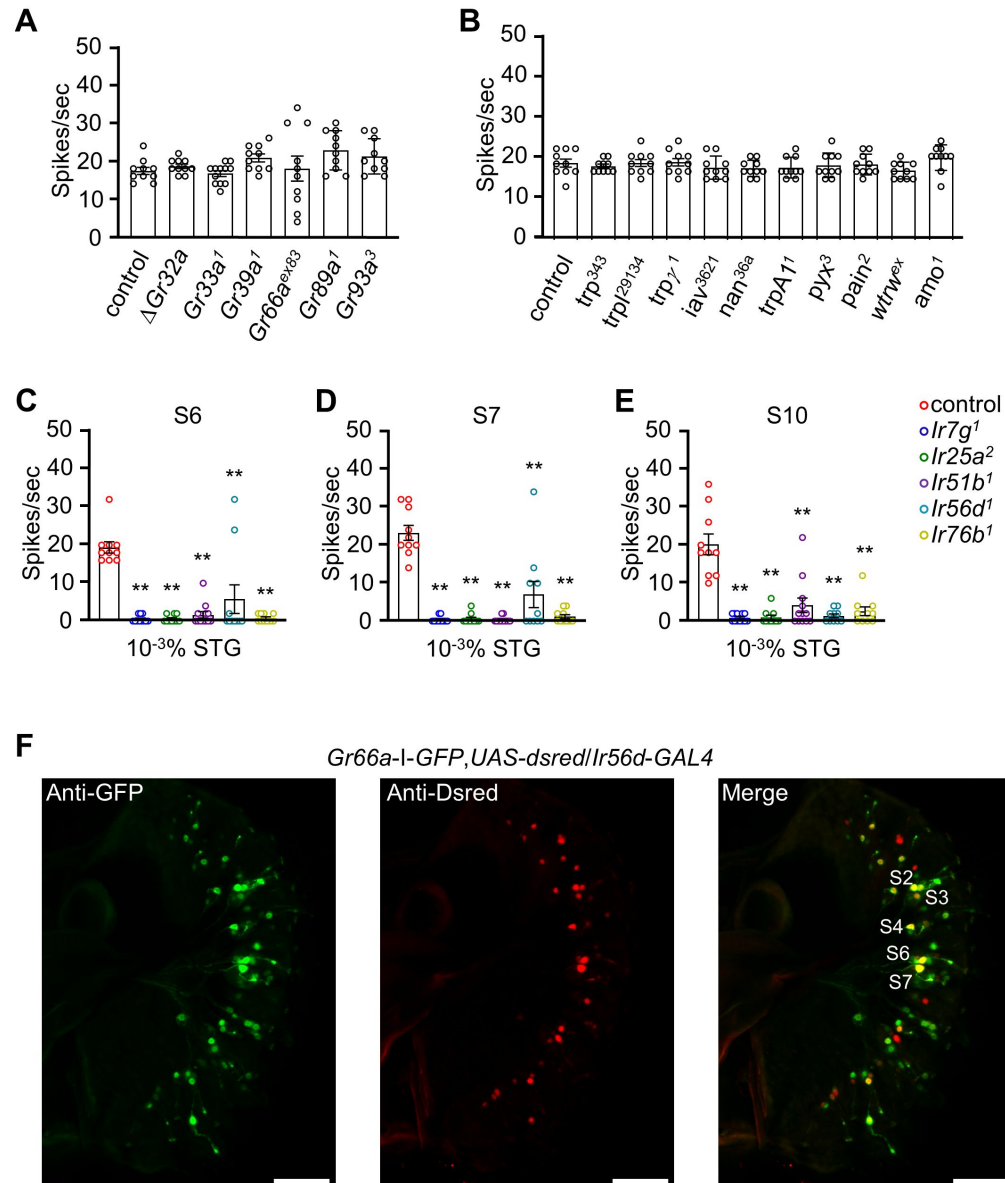


Figure 2—figure supplement 1.

Electrophysiological analysis of different bitter GRs and TRP lines in the presence of 10⁻¹% CHL, with subset expression of *Ir56d* in bitter GRNs.

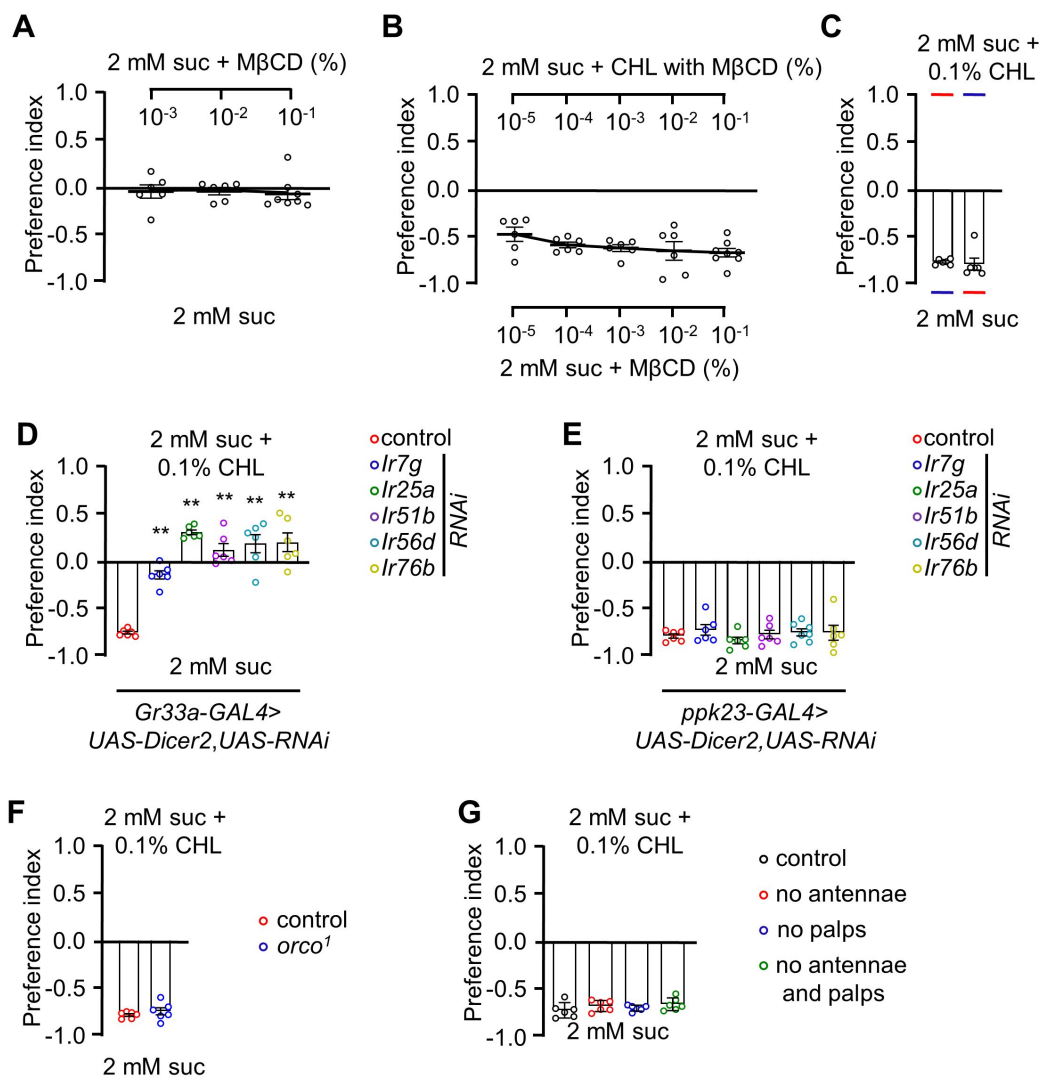


Figure 3—figure supplement 1.

Binary food choice assay with CHL and M β CD.

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Additional information

Author contributions

Roshani Nhuchhen Pradhan: Conceptualization, Investigation, Methodology, Analysis, Writing – Original draft. Craig Montell: Conceptualization, Funding acquisition, Writing – Review & Editing. Youngseok Lee: Conceptualization, Funding acquisition, Supervision, Writing – Review & Editing.

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Reviewer #1 (Public review):

Summary:

Pradhan et al investigated the potential gustatory mechanisms that allow flies to detect cholesterol. They found that flies are indifferent to low cholesterol and avoid high cholesterol. They further showed that the ionotropic receptors Ir7g, Ir51b, and Ir56d are important for the cholesterol sensitivity in bitter neurons. The figures are clear and the behavior result is interesting. However, I have several major comments, especially on the discrepancy of the expression of these Irs with other lab published results, and the confusing finding that the same receptors (Ir7g, Ir51b) have been implicated in the detection of various seemingly unrelated compounds.

Strengths:

The results are very well presented, the figures are clear and well-made, text is easy to follow.

Weaknesses:

(1) Regarding the expression of Ir56d. The reported Ir56d expression pattern contradicts multiple previous studies (Brown et al., 2021 eLife, Figure 6a-c; Sanchez-Alcaniz et al., 2017 Nature Communications, Figure 4e-h; Koh et al., 2014 Neuron, Figure 3b). These studies, using three different driver lines, consistently showed Ir56d expression in sweet-sensing neurons and taste peg neurons. Importantly, Sanchez-Alcaniz et al. demonstrated that Ir56d is not expressed in Gr66a-expressing (bitter) neurons. This discrepancy is critical since Ir56d is identified as the key subunit for cholesterol detection in bitter neurons, and misexpression of Ir7g and Ir51b together is insufficient to confer cholesterol sensitivity (Fig.4b,d). Which Ir56d-GAL4 (and Gr66a-I-GFP) line was used in this study? Is there additional evidence (scRNA sequencing, in-situ hybridization, or immunostaining) supporting Ir56d expression in bitter neurons?

(2) Ir51b has previously been implicated in detecting nitrogenous waste (Dhakal 2021), lactic acid (Pradhan 2024), and amino acids (Aryal 2022), all by the same lab. Additionally, both Ir7g and Ir51b have been implicated in detecting cantharidin, an insect-secreted compound that flies may or may not encounter in the wild, by the same lab. Is Ir51b proposed to be a specific receptor for these chemically distinct compounds or a general multimodal receptor for aversive stimuli? Unlike other multimodal bitter receptors, the expression level of Ir51b is rather low and it's unclear which subset of GRNs express this receptor. The chemical diversity among nitrogenous waste, amino acids, lactic acid, cantharidin, and cholesterol raises questions about the specificity of these receptors and warrants further investigation and at a minimum discussion in this paper. Given the wide and seemingly unrelated sensitivity of Ir51b and Ir7g to these compounds I'm leaning towards the hypothesis that at least some of these is non-specific and ecologically irrelevant without further supporting evidence from the authors.

(3) The Benton lab Ir7g-GAL4 reporter shows no expression in adults. Additionally, two independent labellar RNA sequencing studies (Dweck, 2021 eLife; Bontonou et al., 2024 Nature Communications) failed to detect Ir7g expression in the labellum. This contradicts the authors' previous RT-PCR results (Pradhan 2024 Fig. S4, Journal of Hazardous Materials) showing Ir7g expression in the labellum. Additionally the Benton and Carlson lab Ir51b-GAL4 reporters show no expression in adults as well. Please address these inconsistencies.

(4) The premise that high cholesterol intake is harmful to flies, which makes sensory mechanisms for cholesterol avoidance necessary, is interesting but underdeveloped. Animal sensory systems typically evolve to detect ecologically relevant stimuli with dynamic ranges matching environmental conditions. Given that *Drosophila* primarily consume fruits and plant matter (which contain minimal cholesterol) rather than animal-derived foods (which contain higher cholesterol), the ecological relevance of cholesterol detection requires more thorough discussion. Furthermore, at high concentrations, chemicals often activate multiple receptors beyond those specifically evolved for their detection. If the cholesterol concentrations used in this study substantially exceed those encountered in the fly's natural diet, the observed responses may represent an epiphenomenon rather than an ecologically and ethologically relevant sensory mechanism. What is the cholesterol content in flies' diet and how does that compare to the concentrations used in this paper?

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Reviewer #2 (Public review):

Summary:

In Cholesterol Taste Avoidance in *Drosophila melanogaster*, Pradhan et al. used behavioral and electrophysiological assays to demonstrate that flies can: (1) detect cholesterol through a subset of bitter-sensing gustatory receptor neurons (GRNs) and (2) avoid consuming food with high cholesterol levels. Mechanistically, they identified five members of the IR family as necessary for cholesterol detection in GRNs and for the corresponding avoidance behavior. Ectopic expression experiments further suggested that Ir7g + Ir56d or Ir51b + Ir56d may function as tuning receptors for cholesterol detection, together with the Ir25a and Ir76b co-receptors.

Strengths:

The experimental design of this study was logical and straightforward. Leveraging their expertise in the *Drosophila* taste system, the research team identified the molecular and cellular basis of a previously unrecognized taste category, expanding our understanding of gustation. A key strength of the study was its combination of electrophysiological recordings with behavioral genetic experiments.

Weaknesses:

My primary concern with this study is the lack of a systematic survey of the IRs of interest in the labellum GRNs. Consequently, there is no direct evidence linking the expression of putative cholesterol IRs to the B GRNs in the S6 and S7 sensilla.

Specifically, the authors need to demonstrate that the IR expression pattern explains cholesterol sensitivity in the B GRNs of S6 and S7 sensilla, but not in other sensilla. Instead of providing direct IR expression data for all candidate IRs (as shown for Ir56d in Figure 2-figure supplement 1F), the authors rely on citations from several studies (Lee, Poudel et al. 2018; Dhakal, Sang et al. 2021; Pradhan, Shrestha et al. 2024) to support their claim that Ir7g,

Ir25a, Ir51b, and Ir76b are expressed in B GRNs (Lines 192-194). However, none of these studies provide GAL4 expression or in situ hybridization data to substantiate this claim.

Without a comprehensive IR expression profile for GRNs across all taste sensilla, it is difficult to interpret the ectopic expression results observed in the B GRN of the I9 sensillum or the A GRN of the L-sensillum (Figure 4). It remains equally plausible that other tuning IRs-beyond the co-receptor Ir25a and Ir76b-could interact with the ectopically expressed IRs to confer cholesterol sensitivity, rather than the proposed Ir7g + Ir56d or Ir51b + Ir56d combinations.

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Reviewer #3 (Public review):

Summary:

Whether and how animals can taste cholesterol is not well understood. The study provides evidence that 1) cholesterol activates a subset of bitter-sensing gustatory receptor neurons (GRNs) in the fly labellum, but not other types of GRNs, 2) flies show aversion to high concentrations of cholesterol, and this is mediated by bitter GRNs, and 3) cholesterol avoidance depends on a specific set of ionotropic receptor (IR) subunits acting in bitter GRNs. The claims of the study are supported by electrophysiological recordings, genetic manipulations, and behavioral readouts.

Strengths:

Cholesterol taste has not been well studied, and the paper provides new insight into this question. The authors took a comprehensive and rigorous approach in several different parts of the paper, including screening the responses of all 31 labellar sensilla, screening a large panel of receptor mutants, and performing misexpression experiments with nearly every combination of the 5 IRs identified. The effects of the genetic manipulations are very clear and the results of electrophysiological and behavioral studies match nicely, for the most part. The appropriate controls are performed for all genetic manipulations.

Weaknesses:

The weaknesses of the study, described below, are relatively minor and do not detract from the main conclusions of the paper.

(1) The paper does not state what concentrations of cholesterol are present in *Drosophila*'s natural food sources. Are the authors testing concentrations that are ethologically relevant?

(2) The paper does not state or show whether the expression of IR7g, IR51b, and IR56d is confined to bitter GRNs. Bitter-specific expression of at least some of these receptors would be necessary to explain why bitter GRNs but not sugar GRNs (or other GRN types) normally show cholesterol responses.

(3) The authors only investigated the responses of GRNs in the labellum, but GRN responses in the leg may also contribute to the avoidance of cholesterol feeding. Alternatively, leg GRNs might contribute to cholesterol attraction that is unmasked when bitter GRNs are silenced. In support of this possibility, Ahn et al. (2017) showed that Ir56d functions in sugar GRNs of the leg to promote appetitive responses to fatty acids.

(4) The authors might consider using proboscis extension as an additional readout of taste attraction or aversion, which would help them more directly link the labellar GRN responses to a behavioral readout. Using food ingestion as a readout can conflate the contribution of taste with post-ingestive effects, and the regulation of food ingestion also may involve contributions from GRNs on multiple organs, whereas organ-specific contributions can be

dissociated using proboscis extension. For example, does presenting cholesterol on the proboscis lead to aversive responses in the proboscis extension assay (e.g., suppression of responses to sugar)? Does this aversion switch to attraction when bitter GRNs are silenced, as with the feeding assay?

(5) The authors claim that the cholesterol receptor is composed of IR25a, IR76b, IR56d, and either IR7g or IR51b. While the authors have shown that IR25a and IR76b are each required for cholesterol sensing, they did not show that both are required components of the same receptor complex. If the authors are relying on previous studies to make this assumption, they should state this more clearly. Otherwise, I think further misexpression experiments may be needed where only IR25a or IR76b, but not both, are expressed in GRNs.

<https://doi.org/10.7554/eLife.106256.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

Pradhan et al investigated the potential gustatory mechanisms that allow flies to detect cholesterol. They found that flies are indifferent to low cholesterol and avoid high cholesterol. They further showed that the ionotropic receptors Ir7g, Ir51b, and Ir56d are important for the cholesterol sensitivity in bitter neurons. The figures are clear and the behavior result is interesting. However, I have several major comments, especially on the discrepancy of the expression of these Irs with other lab published results, and the confusing finding that the same receptors (Ir7g, Ir51b) have been implicated in the detection of various seemingly unrelated compounds.

Strengths:

The results are very well presented, the figures are clear and well-made, text is easy to follow.

Weaknesses:

(1) Regarding the expression of Ir56d. The reported Ir56d expression pattern contradicts multiple previous studies (Brown et al., 2021 eLife, Figure 6a-c; Sanchez-Alcaniz et al., 2017 Nature Communications, Figure 4e-h; Koh et al., 2014 Neuron, Figure 3b). These studies, using three different driver lines, consistently showed Ir56d expression in sweet-sensing neurons and taste peg neurons. Importantly, Sanchez-Alcaniz et al. demonstrated that Ir56d is not expressed in Gr66a-expressing (bitter) neurons. This discrepancy is critical since Ir56d is identified as the key subunit for cholesterol detection in bitter neurons, and misexpression of Ir7g and Ir51b together is insufficient to confer cholesterol sensitivity (Fig.4b,d). Which Ir56d-GAL4 (and Gr66a-I-GFP) line was used in this study? Is there additional evidence (scRNA sequencing, in-situ hybridization, or immunostaining) supporting Ir56d expression in bitter neurons?

We agree that the expression pattern of *Ir56d* diverges from two prior reports. The studies by Brown et al. and Koh et al. employed the same *Ir56d-GAL4* driver line, which exhibited expression in sweet-sensing gustatory receptor neurons (GRNs) and taste peg neurons, but not bitter GRNs (the Sanchez-Alcaniz et al. paper did not use an *Ir56d-Gal4*).

In our study, we used a *Ir56d-GAL4* driver line (KDRC:2307) and the *Gr66a-I-GFP* reporter line (Weiss et al., 2011 Neuron). This is a crucial distinction, as differences in the regulatory regions used to generate different driver lines are well known to underlie differences in expression patterns. Our double-labeling experiments revealed co-expression of *Ir56d* with *Gr66a*-positive bitter GRNs specifically within the S6 and S7 sensilla—types previously shown to exhibit strong electrophysiological responses to cholesterol (Figure 2—figure supplement 1F).

We believe this observation is biologically significant and consistent with our functional data. Specifically, targeted expression of *Ir56d* in bitter neurons using the *Gr33a-GAL4* was sufficient to rescue cholesterol avoidance behavior in *Ir56d*¹ mutants (Figure 3G). These results demonstrate that *Ir56d* plays a functional role in bitter GRNs for cholesterol detection. The convergence of genetic, behavioral, and electrophysiological data presented in our study provides compelling support for this previously unappreciated expression pattern and function of *Ir56d*.

(2) Ir51b has previously been implicated in detecting nitrogenous waste (Dhakal 2021), lactic acid (Pradhan 2024), and amino acids (Aryal 2022), all by the same lab. Additionally, both Ir7g and Ir51b have been implicated in detecting cantharidin, an insect-secreted compound that flies may or may not encounter in the wild, by the same lab. Is Ir51b proposed to be a specific receptor for these chemically distinct compounds or a general multimodal receptor for aversive stimuli? Unlike other multimodal bitter receptors, the expression level of Ir51b is rather low and it's unclear which subset of GRNs express this receptor. The chemical diversity among nitrogenous waste, amino acids, lactic acid, cantharidin, and cholesterol raises questions about the specificity of these receptors and warrants further investigation and at a minimum discussion in this paper. Given the wide and seemingly unrelated sensitivity of Ir51b and Ir7g to these compounds I'm leaning towards the hypothesis that at least some of these is non-specific and ecologically irrelevant without further supporting evidence from the authors.

While it is true that *Ir51b* and *Ir7g* are responsive to a range of compounds, they share chemical features such as nitrogen-containing groups, hydrophobicity, or amphipathic structures suggesting that recognition of these chemicals may be mediated by the same or overlapping domains within the receptor complexes. These features could facilitate binding to a structurally diverse yet chemically related groups of aversive ligands.

In the case of cholesterol, while its sterol ring system is distinct from the other compounds, it shares hydrophobic and amphipathic properties that may enable interaction with these receptors via similar structural motifs. Importantly, our data demonstrates that *Ir51b* and *Ir7g* are necessary but not sufficient on their own to confer cholesterol sensitivity, indicating that additional co-factors or receptor subunits are required for full functionality (Figure 4B, D). Furthermore, our dose-response analysis (Figure 3F) shows that *Ir7g* is particularly important at higher cholesterol concentrations, supporting the idea of graded sensitivity rather than indiscriminate activation. This suggests that these receptors may have evolved to recognize cholesterol and its analogs (e.g., phytosterols such as stigmasterol, yet to be tested), which are naturally found in the fly's diet (e.g., yeast and plant-derived matter), as ecologically relevant cues signaling microbial contamination, lipid imbalance, or dietary overconsumption.

We acknowledge the reviewer's concern regarding the relatively low expression levels of *Ir51b* and *Ir7g*. However, we note that low transcript abundance does not necessarily equate to diminished physiological relevance. Finally, we agree that the chemical diversity of ligands associated with *Ir51b* and *Ir7g* warrants deeper investigation, particularly through structure-

function studies aimed at identifying ligand-binding domains and receptor-ligand interactions at atomic resolution.

(3) *The Benton lab Ir7g-GAL4 reporter shows no expression in adults. Additionally, two independent labellar RNA sequencing studies (Dweck, 2021 eLife; Bontonou et al., 2024 Nature Communications) failed to detect Ir7g expression in the labellum. This contradicts the authors' previous RT-PCR results (Pradhan 2024 Fig. S4, Journal of Hazardous Materials) showing Ir7g expression in the labellum. Additionally the Benton and Carlson lab Ir51b-GAL4 reporters show no expression in adults as well. Please address these inconsistencies.*

With respect to *Ir7g*, we acknowledge that the *Ir7g-GAL4* reporter line from the Benton lab does not exhibit detectable expression in adult labella. Furthermore, two independent transcriptomic studies—Dweck et al., 2021 (eLife) and Bontonou et al., 2024 (Nature Communications) also did not detect *Ir7g* transcripts in bulk RNA-seq datasets derived from adult labella. However, our previously published RT-PCR data (Pradhan et al., 2024, Journal of Hazardous Materials, Fig. S4) revealed *Ir7g* expression in labellar tissue, albeit at low levels. Our RT-PCR includes an internal control (*tubulin*) with the same reaction tube with control and the *Ir7g* mutant as a negative control. Therefore, we stand behind the findings that *Ir7g* is expressed in the labellum.

We would like to point out that RT-PCR is more sensitive and better-suited to detect low-abundance transcripts than bulk RNA-seq, which may fail to capture transcripts due to limitations in depth of coverage. Moreover, immunohistochemistry can have limitations in detecting very low expression levels. Costa et al. 2013 (Translational lung cancer research) states that “RNA-Seq technique will not likely replace current RT-PCR methods, but will be complementary depending on the needs and the resources as the results of the RNA-Seq will identify those genes that need to then be examined using RT-PCR methods”.

Similarly, regarding *Ir51b*, while the *GAL4* reporter lines from the Benton and Carlson labs do not show robust adult expression, our RT-PCR and functional data strongly support a role for *Ir51b* in labellar bitter GRNs. Specifically, *Ir51b*¹ mutants display electrophysiological deficits in response to cholesterol (Figure 2A–B), and these defects are rescued by expressing *Ir51b* in *Gr33a*-positive bitter neurons (Figure 3G), providing functional validation of the RT-PCR expression.

(4) *The premise that high cholesterol intake is harmful to flies, which makes sensory mechanisms for cholesterol avoidance necessary, is interesting but underdeveloped. Animal sensory systems typically evolve to detect ecologically relevant stimuli with dynamic ranges matching environmental conditions. Given that Drosophila primarily consume fruits and plant matter (which contain minimal cholesterol) rather than animal-derived foods (which contain higher cholesterol), the ecological relevance of cholesterol detection requires more thorough discussion. Furthermore, at high concentrations, chemicals often activate multiple receptors beyond those specifically evolved for their detection. If the cholesterol concentrations used in this study substantially exceed those encountered in the fly's natural diet, the observed responses may represent an epiphenomenon rather than an ecologically and ethologically relevant sensory mechanism. What is the cholesterol content in flies' diet and how does that compare to the concentrations used in this paper?*

Drosophila melanogaster cannot synthesize sterols *de novo*, and must acquire them from its diet. In natural environments, flies acquire sterols from fermenting fruit, decaying plant matter, and yeast, which contain trace amounts of phytosterols (e.g., stigmasterol, β -sitosterol) and ergosterol. While the exact sterol concentrations in these sources remain

uncharacterized, our behavioral assays used concentrations (0.001–0.01% by weight) that align with the low levels expected in such nutrient-limited ecological niches.

In our study, the cholesterol concentrations tested ranged from 0.001% to 0.1%, thereby spanning both the physiologically relevant and slightly elevated range. Importantly, avoidance behaviors and receptor activation were most prominent at 0.1% cholesterol. While it is true that high chemical concentrations may elicit off-target effects via broad receptor activation, our genetic and electrophysiological data indicate that the observed responses are mediated by specific ionotropic receptors (Ir51b, Ir7g, Ir56d) and not merely generalized chemical stress.

Ecologically, elevated sterol levels may also signal conditions unsuitable for egg-laying or larval development. For example, high levels of cholesterol or other sterols may occur in substrates colonized by pathogenic microbes, decaying animal tissue, or in cases of abnormal microbial fermentation, which could represent a nutritional or microbial hazard. The avoidance of cholesterol may help signal the flies to avoid consuming decaying animal tissue. In this context, sensory detection of excessive cholesterol might serve as a protective function.

Reviewer #2 (Public review):

Summary:

In Cholesterol Taste Avoidance in Drosophila melanogaster, Pradhan et al. used behavioral and electrophysiological assays to demonstrate that flies can: (1) detect cholesterol through a subset of bitter-sensing gustatory receptor neurons (GRNs) and (2) avoid consuming food with high cholesterol levels. Mechanistically, they identified five members of the IR family as necessary for cholesterol detection in GRNs and for the corresponding avoidance behavior. Ectopic expression experiments further suggested that Ir7g + Ir56d or Ir51b + Ir56d may function as tuning receptors for cholesterol detection, together with the Ir25a and Ir76b co-receptors.

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Specifically, the authors need to demonstrate that the IR expression pattern explains cholesterol sensitivity in the B GRNs of S6 and S7 sensilla, but not in other sensilla. Instead of providing direct IR expression data for all candidate IRs (as shown for Ir56d in Figure 2-figure supplement 1F), the authors rely on citations from several studies (Lee, Poudel et al. 2018; Dhakal, Sang et al. 2021; Pradhan, Shrestha et al. 2024) to support their claim that Ir7g, Ir25a, Ir51b, and Ir76b are expressed in B GRNs (Lines 192-194). However, none of these studies provide GAL4 expression or in situ hybridization data to substantiate this claim.

Without a comprehensive IR expression profile for GRNs across all taste sensilla, it is difficult to interpret the ectopic expression results observed in the B GRN of the I9

*sensillum or the A GRN of the L-sensillum (Figure 4). It remains equally plausible that other tuning IRs-beyond the co-receptor *Ir25a* and *Ir76b*-could interact with the ectopically expressed IRs to confer cholesterol sensitivity, rather than the proposed *Ir7g* + *Ir56d* or *Ir51b* + *Ir56d* combinations.*

We provide electrophysiological data demonstrating that the S6 and S7 sensilla respond to cholesterol (Figure 1D). This finding is consistent with the hypothesis that these sensilla harbor the complete receptor complexes necessary for cholesterol detection. In our electrophysiological recordings, only those bitter GRNs that co-express *Ir56d* along with either *Ir7g* or *Ir51b* generate action potentials in response to cholesterol. Other S-type sensilla lacking one or more of these subunits remain unresponsive, reinforcing the idea that these components are necessary for receptor function and sensory coding of cholesterol. Moreover, in the cholesterol-insensitive I9 sensillum (based on our mapping results using electrophysiology), co-expression of either *Ir7g* + *Ir56d* or *Ir51b* + *Ir56d* conferred *de novo* cholesterol sensitivity (Figure 4B). Importantly, no cholesterol response was observed when any of these IRs was expressed alone or when *Ir7g* + *Ir51b* were co-expressed without *Ir56d*. These findings strongly argue against the possibility that endogenous tuning IRs in I9 sensilla (e.g., *Ir25a*, *Ir76b*) are sufficient to generate cholesterol responsiveness.

Furthermore, based on the literature, *Ir25a* and *Ir76b* are endogenously expressed in I- and L-type sensilla. Thus, their presence alone is insufficient for cholesterol responsiveness. These data support the model that cholesterol sensitivity depends on a specific, multi-subunit receptor complex (e.g., *Ir7g* + *Ir25a* + *Ir56d* + *Ir76b* or *Ir51b* + *Ir25a* + *Ir56d* + *Ir76b*).

In conclusion, while we acknowledge that our data do not provide a full anatomical map of IR expression across all sensilla, our results strongly support the idea that cholesterol sensitivity in S6 and S7 sensilla arises from specific combinations of IRs expressed in the B GRNs.

Reviewer #3 (Public review):

Summary:

Whether and how animals can taste cholesterol is not well understood. The study provides evidence that 1) cholesterol activates a subset of bitter-sensing gustatory receptor neurons (GRNs) in the fly labellum, but not other types of GRNs, 2) flies show aversion to high concentrations of cholesterol, and this is mediated by bitter GRNs, and 3) cholesterol avoidance depends on a specific set of ionotropic receptor (IR) subunits acting in bitter GRNs. The claims of the study are supported by electrophysiological recordings, genetic manipulations, and behavioral readouts.

Strengths:

Cholesterol taste has not been well studied, and the paper provides new insight into this question. The authors took a comprehensive and rigorous approach in several different parts of the paper, including screening the responses of all 31 labellar sensilla, screening a large panel of receptor mutants, and performing misexpression experiments with nearly every combination of the 5 IRs identified. The effects of the genetic manipulations are very clear and the results of electrophysiological and behavioral studies match nicely, for the most part. The appropriate controls are performed for all genetic manipulations.

Weaknesses:

The weaknesses of the study, described below, are relatively minor and do not detract from the main conclusions of the paper.

(1) The paper does not state what concentrations of cholesterol are present in Drosophila's natural food sources. Are the authors testing concentrations that are

ethologically Drosophila melanogaster primarily feeds on fermenting fruits and associated microbial communities, especially yeast, which serve as major sources of dietary sterols. These natural food sources are known to contain phytosterols such as stigmasterol and β -sitosterol. One study quantified phytosterols (e.g., stigmasterol, sitosterol) in fruits, reporting concentrations between 1.6–32.6 mg/100 g edible portion (~0.0016–0.0326% wet weight) (Han et al 2008). The range we tested falls within this range. Additionally, ergosterol, the principal sterol in yeast and a structural analog of cholesterol, is present at levels of about 0.005% to 0.02% in yeast-rich environments.

To ensure physiological relevance, we designed our behavioral assays to include a broad concentration range of cholesterol, from $10^{-5}\%$ to $10^{-1}\%$. This spans both physiological levels (0.001–0.01%), which are comparable to those found in the natural diet, and supra-physiological levels (e.g., 0.1%), which exceed natural exposure but help define the threshold for aversive behavior.

Our results demonstrate that flies begin to avoid cholesterol at concentrations $\geq 10^{-3}\%$ more (Figure 3A), which falls within the upper physiological range and may reflect the threshold beyond which cholesterol or related sterols become deleterious. At these higher concentrations, excess sterols may disrupt membrane fluidity, interfere with hormone signaling, or promote microbial overgrowth—all of which could compromise fly health.

(2) The paper does not state or show whether the expression of IR7g, IR51b, and IR56d is confined to bitter GRNs. Bitter-specific expression of at least some of these receptors would be necessary to explain why bitter GRNs but not sugar GRNs (or other GRN types) normally show cholesterol responses.

We show the *Ir56d-Gal4* is co-expressed with *Gr66a-GFP* in S6/S7 sensilla, indicating that it is expressed in bitter GRNs (Figure 2—figure supplement 1F). In the case of *Ir7g* and *Ir51b*, there are no reporters or antibodies to address expression. However, previously they have been shown to be expressed in bitter GRNs using RT-PCR (Dhakal et al. 2021, Communications Biology; Pradhan et al. 2024, Journal of Hazardous Materials). In addition, we provide functional evidence that bitter GRNs are required for the cholesterol response since silencing bitter GRNs abolishes cholesterol-induced action potentials (Figure 1E–F). Moreover, we showed that we could rescue the *Ir7g¹*, *Ir51b¹* and *Ir56d¹* mutant phenotypes only when we expressed the cognate transgenes in bitter GRNs using the *Gr33a-GAL4* (Figure 3G). Thus, while *Ir7g/Ir51b* are not exclusive to bitter GRNs, their functional role in cholesterol detection is bitter-GRN-specific.

(3) The authors only investigated the responses of GRNs in the labellum, but GRN responses in the leg may also contribute to the avoidance of cholesterol feeding. Alternatively, leg GRNs might contribute to cholesterol attraction that is unmasked when bitter GRNs are silenced. In support of this possibility, Ahn et al. (2017) showed that Ir56d functions in sugar GRNs of the leg to promote appetitive responses to fatty acids.

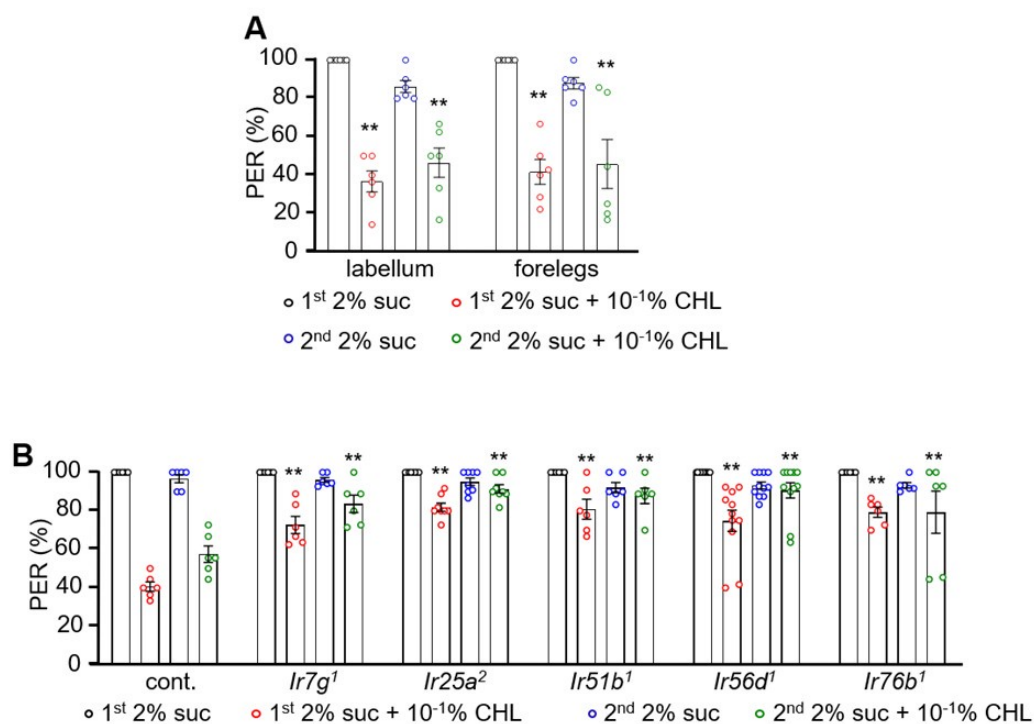
This is an interesting idea. Indeed, when bitter GRNs are hyperpolarized, the flies exhibit a strong attraction to cholesterol. Nevertheless, the cellular basis for cholesterol attraction and whether it is mediated by GRNs in the legs will require a future investigation.

(4) The authors might consider using proboscis extension as an additional readout of taste attraction or aversion, which would help them more directly link the labellar GRN responses to a behavioral readout. Using food ingestion as a readout can conflate the contribution of taste with post-ingestive effects, and the regulation of food ingestion also may involve contributions from GRNs on multiple organs, whereas organ-specific contributions can be dissociated using proboscis extension. For example, does

presenting cholesterol on the proboscis lead to aversive responses in the proboscis extension assay (e.g., suppression of responses to sugar)? Does this aversion switch to attraction when bitter GRNs are silenced, as with the feeding assay?

We thank the reviewer for the suggestion regarding the use of the proboscis extension reflex (PER) assay to strengthen the link between labellar GRN activity and behavioral responses to cholesterol.

Author response image 1.



Our PER assay results shown above indicate that cholesterol presentation on the labellum or forelegs leads to an aversive response, as evidenced by a significant reduction in proboscis extension when compared to control stimuli (Author response image 1A). 2% sucrose or 2% sucrose with 10⁻¹% cholesterol was applied to labellum or forelegs and the percent PER was recorded. n=6. Data were compared using single-factor ANOVA coupled with Scheffe's post-hoc test. Statistical significance was compared with the control. Means ± SEMs. **p<0.01). This finding supports the idea that cholesterol is detected by labellar and leg GRNs and elicits behavioral avoidance. In contrast, sucrose stimulation robustly induces proboscis extension, as expected for an appetitive stimulus. We confirmed the defects of due to each *Ir* mutant by presenting the stimuli to the labellum (Author response image 1B). Together, these PER results provide a more direct behavioral correlate of labellar and leg GRN activation and reinforce our conclusion that cholesterol is sensed as an aversive tastant through the labellar bitter GRNs.

(5) The authors claim that the cholesterol receptor is composed of *IR25a*, *IR76b*, *IR56d*, and either *IR7g* or *IR51b*. While the authors have shown that *IR25a* and *IR76b* are each required for cholesterol sensing, they did not show that both are required components of the same receptor complex. If the authors are relying on previous studies to make this assumption, they should state this more clearly. Otherwise, I think further misexpression

experiments may be needed where only *IR25a* or *IR76b*, but not both, are expressed in GRNs.

In our study, we relied on prior work demonstrating that *Ir25a* and *Ir76b* function as broadly required co-receptors in most IR-dependent chemosensory pathways (Ganguly et al., 2017; Lee et al., 2018). These studies showed that *Ir25a* and *Ir76b* are co-expressed in many GRNs across multiple taste modalities. Functional IR complexes often fail to form or signal properly in the absence of these co-receptors. Thus, it is widely accepted in the field that *Ir25a* and *Ir76b* function together as a core heteromeric scaffold for diverse IR complexes, akin to co-receptors in other ionotropic glutamate receptor families. We state that while *Ir25a* and *Ir76b* are presumed co-receptors in the cholesterol receptor complex based on their conserved roles, their direct physical interaction with *Ir7g*, *Ir51b*, and *Ir56d* remains to be demonstrated.

In support of this model, we note that in our ectopic expression experiments using I9 sensilla, which endogenously express *Ir25a* and *Ir76b*, introduction of either *Ir7g* + *Ir56d* or *Ir51b* + *Ir56d* was sufficient to confer cholesterol sensitivity (Figure 4B). We obtained a similar result in L6 sensilla (Figure 4D), which also endogenously express *Ir25a* and *Ir76b*. These findings imply that both co-receptors are already present in these sensilla and are likely part of the functional complex. However, we agree that we have not directly tested the requirement for both co-receptors in a minimal reconstitution context, such as expressing only *Ir25a* or *Ir76b* alongside tuning IRs in an otherwise null background. Such an experiment would indeed provide more direct evidence of their joint requirement in the receptor complex. Future studies, including heterologous expression experiments, will be necessary to define the cholesterol-receptor complexes.

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